Know Your Geography Know Your India !

INDIA'S SOIL, VEGETATION & RAINFALL

LAND • LIFE • WATER — THE FOUNDATION OF INDIA

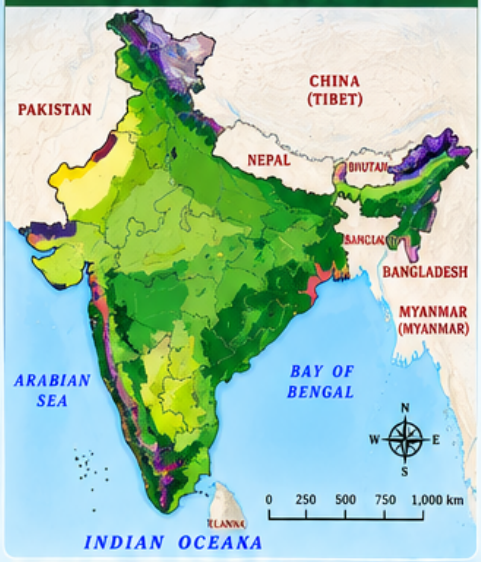
Same Land Different Shades Same Rain Different Life !

MAJOR SOIL TYPES OF INDIA

Soil Shapes Agriculture, Economy and Life

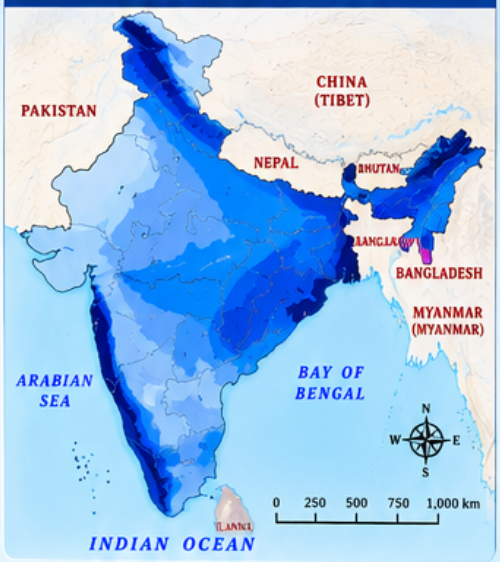
NATURAL VEGETATION OF INDIA

Diverse Flora for a Diverse Land



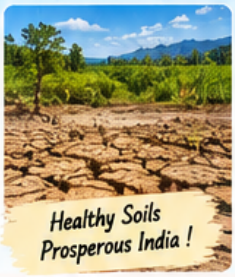
AVERAGE ANNUAL RAINFALL OF INDIA

The Monsoon that Unites Us



SOIL LEGEND

- Alluvial Soil
- Black Soil (Regur)
- Red Soil
- Laterite Soil
- Arid (Desert) Soil
- Forest & Mountain Soil
- Saline & Alkaline Soil
- Peaty & Marshy Soil



Healthy Soils Prosperous India !

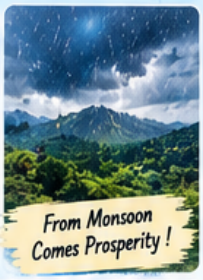
VEGETATION LEGEND

- Tropical Evergreen Forest
- Tropical Deciduous Forest
- Thorn Forest & Scrub
- Montane Forest
- Mangrove Forest
- Grasslands
- Alpine Vegetation
- Littoral & Swamp (Coastal)

Forests Breathe Life into India !

RAINFALL LEGEND (Annual)

- > 400 cm (Very Heavy)
- 200 – 400 cm (Heavy)
- 100 – 200 cm (Moderate)
- 50 – 100 cm (Low)
- < 50 cm (Very Low)



From Monsoon Comes Prosperity !

MAJOR SOIL TYPES – KEY FACTS

Soil Type	Major Areas (States/Regions)	Key Features	Main Crops
Alluvial	Indo-Gangetic Plain, River valleys, Coastal areas	Most fertile, rich in potash	Rice, Wheat, Sugarcane, Jute
Black (Regur)	Deccan Plateau (MH, MP, TN, AP, KtK)	Retains moisture, made of lava	Cotton, Soybean, Sugarcane
Red	Peninsular India (TN, AP, KtK, Odh, Chh)	Rich in iron oxide, less fertile	Millets, Pulses, Groundnut
Laterite	Western & Eastern Ghats (Kerala, KtK, TN, Odh)	Formed in high rainfall areas	Tea, Coffee, Cashew, Rubber
Arid (Desert)	Rajasthan, Gujarat	Sandy, low humus, high salt content	Bajra, Jowar, Pulses (with irrigation)
Forest & Mountain	Himalayas, NE India	Rich in humus	Tea, Spices, Horticulture
Saline & Alkaline	Rajasthan, Haryana, UP, Gujarat (Rann)	High salt content (usar)	Limited cultivation (with reclamation)
Peaty & Marshy	Sundarbans, Kerala, North Bihar, Coastal areas	High organic matter	Rice (with reclamation)

NATURAL VEGETATION – KEY FACTS

Vegetation Type	Major Areas (Regions/States)	Key Features	Example Flora
Tropical Evergreen	Western Ghats, NE India, Andaman & Nicobar	Dense, multi-layered, > 200 cm rainfall	Rosewood, Ebony, Mahogany
Tropical Deciduous (Monsoon)	Central & Eastern India, Foothills of Himalayas	Most widespread, shed leaves in dry season	Teak, Sal, Shisham
Thorn Forest & Scrub	Rajasthan, Gujarat, Deccan (dry areas)	Drought-resistant plants, sparse	Acacia, Babul, Khejri
Montane Forest	Himalayas (J&K to Arunachal)	Varies with altitude	Pine, Deodar, Fir, Spruce
Mangrove Forest	Sundarbans (WB), Andaman & Nicobar, Coastal areas	Tidal, saline, deltaic regions	Sundari, Rhizophora
Grasslands	Terai, Bhabar, Deccan Plateau, Semi-arid regions	Dominated by grasses, used for grazing	Elephant grass, Bamboo (in some)
Alpine Vegetation	Higher Himalayas (> 3,600 m)	Low-growing plants, short growing season	Mosses, Lichens, Alpine herbs
Littoral & Swamp	Coastal areas, Deltas, Backwaters	Salt-tolerant, marshy areas	Coconut, Palm, Screw pine

RAINFALL – KEY FACTS

Rainfall Range (Annual)	Major Areas	Key Features
> 400 cm (Very Heavy)	Mawsynram, Cherrapunji (Meghalaya), Western Ghats (Windward)	Highest rainfall in the world (Mawsynram)
200 – 400 cm (Heavy)	NE India, Western Ghats, Andaman & Nicobar	High rainfall due to relief & monsoon winds
100 – 200 cm (Moderate)	Eastern India, Coastal Karnataka, Sub-Himalayan West Bengal	Good for agriculture
50 – 100 cm (Low)	Interior Peninsular India, Parts of Gujarat, MP	Rain shadow regions
< 50 cm (Very Low)	Thar Desert (Rajasthan), Ladakh, parts of Gujarat	Arid climate, very low rainfall

QUICK FLOW CHARTS & MEMORY TRICKS

Soil Order (N → S)

Alluvial (Indo-Gangetic Plain)
↓
Arid (Rajasthan)
↓
Black (Deccan)
↓
Red (Peninsular)
↓
Laterite (High Rainfall)
↓
Forest (Himalayas)

Vegetation (Rainfall Based)

- > 200 cm → Evergreen Forest
- 100 – 200 cm → Deciduous Forest
- 50 – 100 cm → Thorn/Scrub
- Coastal → Mangroves
- High Altitude → Alpine Vegetation
- Grasslands → Terai & Deccan

Rainfall (West → East)

- VERY HIGH — NE India (Mawsynram)
- HIGH — Western Ghats (Windward)
- MODERATE — Indo-Gangetic Plain
- LOW — Deccan Plateau
- VERY LOW — Rajasthan (Thar)

TOP FACTS FOR SSC CGL

- ✅ Alluvial soil is the most fertile.
- ✅ Black soil is ideal for cotton.
- ✅ Laterite soil is formed in high rainfall areas.
- ✅ Red soil is rich in iron oxide.
- ✅ Thar Desert has arid soil and low rainfall.
- ✅ Western Ghats (windward) receive heavy rainfall.
- ✅ Mawsynram (Meghalaya) is the wettest place in the world.
- ✅ Tropical deciduous forest is the most widespread.
- ✅ Sundarbans has mangrove forests.
- ✅ Alpine vegetation is found above 3,600 m.
- ✅ Vegetation and soil are closely related to climate and relief.
- ✅ Rainfall decreases from east to west.
- ✅ Use map-based questions to identify regions.

ONE INDIA – MANY SHADES

Different Soils
Different Forests
Different Rains
One Incredible India !

"Where Soil Nurtures, Forests Breathe, and Rains Bring Life — There Stands a Stronger India."

GEOGRAPHY | SSC CGL | LEARN | PRACTICE | SUCCEED

"Nature's Diversity Our Nation's Strength !"

Rocks Form Landscapes Landscapes Shape Lives !

THE LITHOSPHERE OF THE EARTH

SOLID GROUND • DYNAMIC PLANET • SHAPING OUR TOMORROW

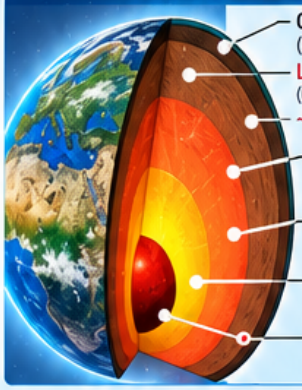
Same Earth Different Landforms One Home Let's Protect It !

WHAT IS LITHOSPHERE?

- The lithosphere is the solid, outermost layer of the Earth, consisting of the crust and the uppermost part of the mantle.
- It is rigid and broken into large pieces called tectonic plates.
- **Thickness: ~ 50–200 km**
  - Oceanic regions: ~ 50–100 km
  - Continental regions: ~ 100–200 km

"The lithosphere is Earth's solid shell, where land, life and civilization exist."

STRUCTURE OF THE EARTH
(Where Lithosphere Lies)



LAYERS COMPARISON

Layer	Depth (km)	Composition	Nature
Crust	5 – 70	Granitic (continental) Basaltic (oceanic)	Solid, rigid
Upper Mantle (part of lithosphere)	Upto ~200	Peridotite (rich in Mg, Fe)	Rigid
Lithosphere	~50 – 200	Crust + Uppermost Mantle	Rigid, broken into plates
Asthenosphere	~200 – 700	Semi-molten rock (magma)	Plastic, flowable
Lower Mantle	700 – 2,900	Silicates (Mg, Fe)	Solid (high pressure)
Outer Core	2,900 – 5,150	Iron, Nickel (liquid)	Liquid
Inner Core	5,150 – 6,371	Iron, Nickel (solid)	Solid

TYPES OF CRUST

Continental Crust

- Thickness: 30–70 km
- Composition: Granitic (Si + Al)
- Less dense (~2.7 g/cm³)
- Older (up to 4 billion years)
- Forms continents

Oceanic Crust

- Thickness: 5–10 km
- Composition: Basaltic (Si + Mg)
- More dense (~3.0 g/cm³)
- Younger (< 200 million years)
- Forms ocean floors

TECTONIC PLATES OF THE WORLD

MAJOR TECTONIC PLATES
(Area-wise)

Plate	Area (million km²)	Key Regions
Pacific	103.3	Major oceans, Ring of Fire
Eurasian	67.8	Europe, Asia
African	61.3	Africa, Atlantic
North American	75.9	N. America, Atlantic, Arctic
South American	43.6	S. America
Antarctic	60.9	Antarctica, Southern Ocean
Indo-Australian	58.9	India, Australia, Indian Ocean
Nazca	15.6	Eastern Pacific (off S. America)
Cocos	2.9	Eastern Pacific (Central America)
Philippine	5.5	Philippines, W. Pacific
Arabian	5.0	Arabian Peninsula
Caribbean	3.3	Caribbean Sea
Scotia	1.6	Southern Ocean

LANDFORMS OF THE LITHOSPHERE

Mountains
(Uplifted land)

Plateaus
(High flat land)

Plains
(Flat land)

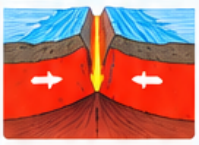
Deserts
(Arid land)

Hills
(Lower elevations)

Volcanoes
(Constructive landforms)

PLATE BOUNDARIES & LANDFORMS

1. Divergent Boundary
(Plates move apart)

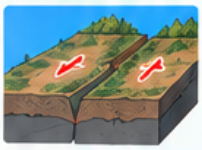


- New crust is formed
- Mid-ocean ridges, rift valleys
- Example: Mid-Atlantic Ridge, East African Rift

2. Convergent Boundary
(Plates collide)

- Destruction of crust (subduction)
- Mountains, trenches, volcanoes
- Example: Himalayas (India–Eurasia), Andes (Nazca–S. America)

3. Transform Boundary
(Plates slide past)



- No crust is created or destroyed
- Earthquakes common
- Example: San Andreas Fault (USA)

RING OF FIRE

- A horseshoe-shaped zone of active volcanoes and earthquakes.
- Caused by subduction of oceanic plates.
- Around 75% of the world's active volcanoes are in this region.

KEY FACTS (QUICK REVISION)

- Lithosphere = Crust + Uppermost Mantle (~50–200 km).
- Broken into tectonic plates.
- Continental crust is thicker and less dense (granitic).
- Oceanic crust is thinner and more dense (basaltic).
- Plates move due to convection currents in the mantle (asthenosphere).
- Major landforms (mountains, volcanoes, trenches) are formed at plate boundaries.
- Ring of Fire is the most active seismic and volcanic zone.
- Himalayas are formed by collision between Indian and Eurasian plates.
- East African Rift is an example of a divergent boundary.
- San Andreas Fault is an example of a transform boundary.

FAMOUS EXAMPLES

Feature	Type	Location	Formed By
Himalayas	Mountain	India	Convergent (India–Eurasia)
Andes	Mountain	South America	Convergent (Nazca–S. America)
Mid-Atlantic Ridge	Oceanic Ridge	Atlantic Ocean	Divergent
East African Rift	Rift Valley	Africa	Divergent
Mariana Trench	Ocean Trench	Pacific Ocean	Convergent
San Andreas Fault	Fault Zone	USA	Transform
Deccan Plateau	Plateau	India	Volcanic (Hotspot)
Thar Desert	Desert	India	Arid landform
Iceland	Volcanic Island	N. Atlantic	Divergent

DID YOU KNOW ?

- The lithosphere "floats" on the semi-molten asthenosphere.
- Earth has 7 major and several minor tectonic plates.
- The Pacific Plate is the largest and most active.
- The oldest rocks are found in the continental crust (up to 4 billion years).
- The oceanic crust is continuously recycled through subduction.
- Most earthquakes and volcanoes occur at plate boundaries.
- The lithosphere is not a continuous layer, but a mosaic of moving plates.

Strong Lands Support Bright Futures

Our Planet Our Responsibility

EXPLORE | LEARN | RESPECT | CONSERVE

Same Air
Same Sky
One Earth
One Humanity !

THE ATMOSPHERE OF THE WORLD

A BLANKET OF GASES — SUPPORTING LIFE, CONNECTING THE PLANET

Breathe
Learn
Respect
Protect !

LAYERS OF THE ATMOSPHERE

Altitude (km)

EXOSPHERE (700 – 10,000 km)
- Outermost layer
- Very thin air
- Merges with outer space
- Satellites orbit here

Satellites

700
600
500

THERMOSPHERE (80 – 700 km)
- Very low air density
- High temperature (500°C – 2000°C)
- Auroras occur here
- ISS orbits (~400 km)

Aurora

400
300
200
100

MESOSPHERE (50 – 80 km)
- Temperature decreases
- Meteors burn here
- Coldest layer (–90°C)

Meteors

50
52

STRATOSPHERE (12 – 50 km)
- Contains Ozone Layer
- Temperature increases
- Stable layer
- Jet aircraft fly here (upper part)

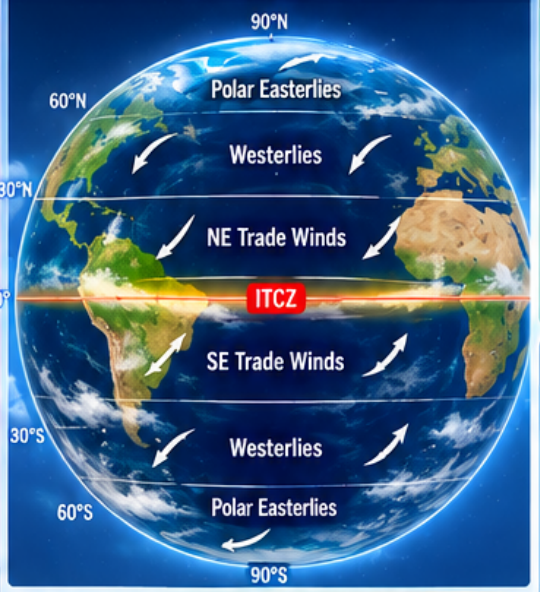
Jet Aircraft

12
12
0

TROPOSPHERE (0 – 12 km)
- Densest layer (75% air)
- Weather phenomena occur here
- Temperature decreases with height (6.5°C per km)

Clouds & Weather

GLOBAL ATMOSPHERIC CIRCULATION



Atmosphere connects every corner of the Earth — what happens in one region, affects all.

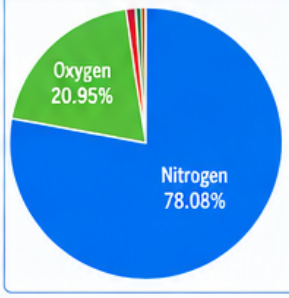
MAJOR PRESSURE BELTS OF THE WORLD

Pressure Belt	Location (Latitude)	Type	Key Features
Equatorial Low (ITCZ)	0°	Low	Rising air, heavy rainfall, calm winds (Doldrums)
Subtropical High	~30° N & S	High	Descending air, clear skies, deserts
Subpolar Low	~60° N & S	Low	Meeting of warm & cold air, cyclonic activity
Polar High	90° N & S	High	Very cold, descending air, dry

PERMANENT WIND BELTS

Wind Belt	Location	Direction	Key Features
Trade Winds	0° – 30°	NE (N Hem) SE (S Hem)	Blow towards equator, steady & reliable
Westerlies	30° – 60°	W → E	Dominant in mid-latitudes, bring rain (esp. to western coasts)
Polar Easterlies	60° – 90°	E → W	Cold, dry winds from polar regions

COMPOSITION OF DRY AIR (Near Surface)

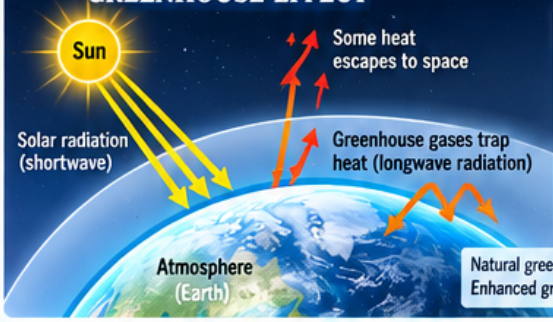


Gas	Percentage
Nitrogen (N_2)	78.08%
Oxygen (O_2)	20.95%
Argon (Ar)	0.93%
Carbon Dioxide (CO_2)	0.04%
Neon (Ne)	0.0018%
Helium (He)	0.0005%
Methane (CH_4)	0.0002%
Krypton (Kr)	0.0001%
Hydrogen (H_2)	0.00005%
Others	0.0013%

VARIABLE COMPONENTS OF AIR

Component	Details
Water Vapour	0–4% (varies by place & time), responsible for clouds, rain, humidity
Carbon Dioxide	~0.04% (increasing due to human activity)
Dust Particles	Affect climate, help in condensation
Pollutants	SO_2, NO_x, O_3 (tropospheric), etc.
Ozone (O_3)	Very small amount in troposphere, high concentration in stratosphere
Aerosols	Natural (volcanic, sea salt) & human-made (smoke, industrial)

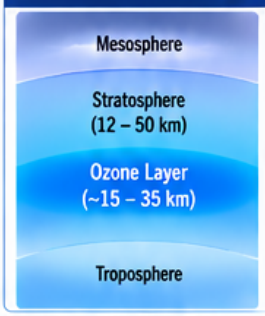
GREENHOUSE EFFECT



Major Greenhouse Gases
- Water Vapour (H_2O)
- Carbon Dioxide (CO_2)
- Methane (CH_4)
- Nitrous Oxide (N_2O)
- Ozone (O_3)
- Chlorofluorocarbons (CFCs)

Natural greenhouse effect keeps Earth warm (~15°C). Enhanced greenhouse effect leads to **Global Warming.**

OZONE LAYER



- ✅ Located in Stratosphere (15–35 km)
- ✅ Absorbs harmful UV radiation (from the Sun)
- ✅ Protects life on Earth
- ✅ Depletion caused by CFCs, halons, etc.
- ✅ Largest depletion: over Antarctica (Ozone Hole)
- ✅ Recovery expected by mid-21st century due to Montreal Protocol (1987)

IMPORTANT JET STREAMS

Jet Stream	Location	Direction	Key Features
Polar Jet Stream	~60° N & S	W → E	Stronger, associated with cold air
Subtropical Jet Stream	~30° N & S	W → E	Weaker, associated with warm air
Tropical Easterly Jet (TEJ)	~10° N (summer)	E → W	Influences Indian monsoon
Somali Jet	Over Arabian Sea (summer)	SW → NE	Helps in onset of Indian monsoon

KEY FACTS (SSC & OTHER EXAMS)

- ✅ Troposphere contains 75% of total air.
- ✅ Temperature decreases with height in troposphere (~6.5°C per km).
- ✅ Ozone layer is in stratosphere.
- ✅ Coldest layer – Mesosphere.
- ✅ Hottest layer – Thermosphere.
- ✅ ITCZ causes heavy rainfall.
- ✅ Westerlies bring rain to Western Europe.
- ✅ Trade winds are responsible for tropical climate.
- ✅ Subtropical high pressure belts are associated with deserts.
- ✅ Jet streams influence weather and air travel.
- ✅ CFCs deplete ozone layer (Montreal Protocol).
- ✅ Greenhouse gases cause global warming.

DO YOU KNOW ?

- Earth's atmosphere extends up to ~10,000 km.
- 99% of atmospheric mass is within 32 km.
- Air pressure decreases with altitude.
- Mount Everest is still within the troposphere.
- Auroras occur in the thermosphere.
- Most weather and climate events occur in troposphere.
- Deserts are located near 30° N & S (subtropical high).
- Doldrums are near the equator (ITCZ).
- Polar regions have high pressure due to very cold air.
- Jet streams can reach speeds over 300 km/h.

A Healthy Atmosphere
A Healthy Planet
A Brighter Tomorrow !

"Clean Air Today for Generations Tomorrow !"

Same Planet | Same Atmosphere | Shared Future

Mountains Connect People, Cultures and Opportunities !

IMPORTANT PASSES OF INDIA

Location | Connects | Altitude | Strategic Importance | For SSC CGL & Other Exams

Know the Passes Know the Power of India !

KEY FACTS (SSC CGL)

- ✅ Majority of passes in Himalayas.
- ✅ Connect India with neighbouring countries.
- ✅ Important for trade, defence and strategic movement.
- ✅ Often used in ancient trade routes.
- ✅ Many passes remain closed in winter due to snow.
- ✅ Frequently asked in SSC CGL, UPSC, State PSCs.

Khardung La
World's highest motorable pass (5,359 m)

Nathu La
Strategic pass to China

Rohtang Pass
Gateway to Lahaul & Spiti

Zoji La
Connects Kashmir with Ladakh

Pal Ghat
Important pass in Western Ghats

PASSES IN NORTHERN & NORTH-EASTERN INDIA (HIMALAYAS)

S. No.	Pass	State/UT	Connects	Altitude (m)
1	Karakoram Pass	Ladakh	India – China	5,540
2	Aghil Pass	Ladakh	India – China (Xinjiang)	4,822
3	Mintaka Pass	Ladakh	India – China	4,693
4	Parpik Pass	Ladakh	India – China	4,370
5	Khardung La	Ladakh	Leh – Nubra Valley	5,359
6	Chang La	Ladakh	Leh – Pangong Tso	5,360
7	Zoji La	J&K	Srinagar – Leh	3,528
8	Burzil Pass	J&K	Uri – Astore (PoK)	4,267
9	Banihal Pass	J&K	Jammu – Kashmir Valley	2,832
10	Fotu La	Ladakh	Leh – Kargil – Srinagar	4,108
11	Rohtang Pass	Himachal Pradesh	Kullu – Lahaul & Spiti	3,978
12	Shipki La	Himachal Pradesh	India – China (Tibet)	3,930
13	Mana Pass	Uttarakhand	India – China (Tibet)	5,610
14	Niti Pass	Uttarakhand	India – China (Tibet)	5,070
15	Lipu Lekh	Uttarakhand	India – China (Tibet)	5,334
16	Nathu La	Sikkim	India – China (Tibet)	4,310
17	Jelep La	Sikkim	India – China (Tibet)	4,267
18	Sela Pass	Arunachal Pradesh	Tawang – Dirang	4,170
19	Bomdi La	Arunachal Pradesh	Tawang – West Kameng	4,170
20	Yanggyap Pass	Arunachal Pradesh	India – China (Tibet)	4,730
21	Diphu Pass	Arunachal Pradesh	India – Myanmar	3,530

PASSES IN NORTH-WESTERN INDIA

S. No.	Pass	State/UT	Connects	Altitude (m)
1	Khyber Pass	Historic, now in Pakistan)	India – Afghanistan	~1,070
2	Mujeeb Pass	J&K (PoK)	India – Pakistan	3,297

PASSES IN WESTERN INDIA

S. No.	Pass	State/UT	Connects	Altitude (m)
1	Thal Ghat	Maharashtra	Nashik – Kasara	701
2	Bhor Ghat	Maharashtra	Mumbai – Pune	550

PASSES IN SOUTHERN INDIA

S. No.	Pass	State/UT	Connects	Altitude (m)
1	Pal Ghat	Kerala – Tamil Nadu	Kerala – Coimbatore	140

Ancient Routes, Modern Strength
Passes Shape India's Destiny !

"Crossing Passes, Connecting Horizons"

GEOGRAPHY | SSC CGL | LEARN | PRACTICE | SUCCEED

EARTHQUAKES

THE EARTH MOVES, BUT KNOWLEDGE KEEPS US SAFE !

SHAKING EARTH, SHAPING TOMORROW

UNDERSTAND • PREPARE • STAY SAFE

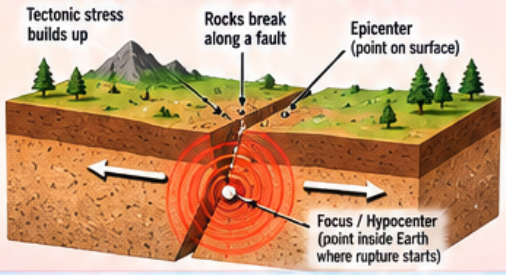
Safer Communities Stronger Tomorrow !

WHAT IS AN EARTHQUAKE?

- Sudden shaking of the Earth's surface caused by the rapid release of energy in the lithosphere.
- Energy is released in the form of seismic waves.
- Occurs due to movement of tectonic plates, volcanic activity, or human activities.

"Earthquakes don't kill people, unsafe buildings do."

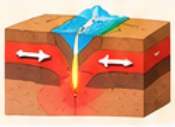
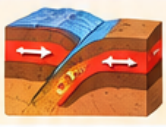
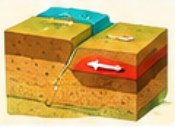
HOW EARTHQUAKES OCCUR?



TYPES OF EARTHQUAKES

Tectonic		Most common (≈ 90%). Caused by movement of tectonic plates.
Volcanic		Caused by movement of magma. Usually smaller in magnitude.
Collapse		Due to collapse of underground cavities (mines, caves).
Induced		Caused by human activities like dams, mining, fluid injection, nuclear tests.

TECTONIC PLATE BOUNDARIES

Divergent (Plates move apart)	Convergent (Plates collide)	Transform (Plates silde past)
		
Magma rises, new crust is formed.	One plate goes beneath another (subduction). Most powerful earthquakes.	Plates move horizontally past each other. Example: San Andreas Fault.

SEISMIC WAVES

Type	Speed	Nature	Effects
P-waves (Primary)	Fastest (5–14 km/s)	Compressional (Longitudinal)	Travel through solids, liquids & gases (least destructive)
S-waves (Secondary)	Slower (3–8 km/s)	Shear (Transverse)	Travel only through solids (more destructive)
Surface Waves (Love & Rayleigh)	Slowest (2–4 km/s)	Move along Earth's surface	Most destructive

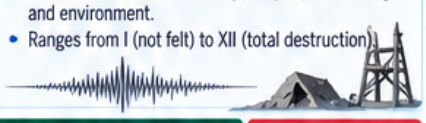
MEASURING EARTHQUAKES

1. Magnitude (Richter Scale / Moment Magnitude)

- Measures energy released.
- Logarithmic scale (each unit = 10 times more energy).
- Now, Moment Magnitude Scale (M_W) is widely used.

2. Intensity (Modified Mercalli Scale)

- Measures effects and damage on people, buildings and environment.
- Ranges from I (not felt) to XII (total destruction).



MAJOR EARTHQUAKE ZONES OF THE WORLD

Around 90% of the world's earthquakes occur in the Circum-Pacific Belt (Ring of Fire).

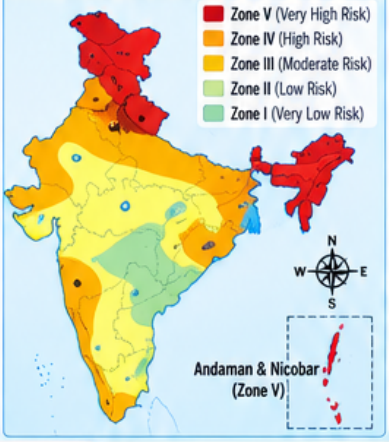
RING OF FIRE

- ✅ Horseshoe-shaped zone surrounding the Pacific Ocean.
- ✅ Site of frequent earthquakes, volcanoes and tsunamis.
- ✅ Includes coastal regions of: North & South America, Japan, Philippines, Indonesia, New Zealand etc.
- ✅ Caused by subduction of oceanic plates.

"Most Active, Most Powerful A Defining Zone on Earth !"

EARTHQUAKE ZONES OF INDIA

(Seismic Zonation Map - IS 1893:2016)



HIGH RISK AREAS IN INDIA (Zone V)

- Jammu & Kashmir
- Ladakh
- Himachal Pradesh
- Uttarakhand
- North East India (Arunachal Pradesh, Assam, etc.)
- Andaman & Nicobar Islands
- Parts of Gujarat (Kachchh)

"High Risk Demands High Readiness !"

MAJOR EARTHQUAKES IN THE WORLD

Year	Location	Magnitude	Key Facts
1960	Valdivia, Chile	9.5	Strongest ever recorded
1964	Alaska, USA	9.2	Great Alaska Earthquake
2004	Sumatra, Indonesia	9.1	Triggered a massive tsunami (Indian Ocean)
2011	Tohoku, Japan	9.0	Caused tsunami and Fukushima nuclear disaster
2010	Haiti	7.0	Massive destruction
2023	Turkey–Syria	7.8	Widespread devastation

MAJOR EARTHQUAKES IN INDIA

Year	Location	Magnitude	Key Facts
1897	Shillong (Assam)	8.1	Great Assam Earthquake
1905	Kangra (Himachal)	7.8	Heavy destruction
1934	Bihar–Nepal	8.1	Widespread damage
2001	Bhuj (Gujarat)	7.7	Massive destruction
2011	Sikkim	6.9	Severe impact in NE India
2015	Nepal (affecting India)	7.8	Heavy damage in Uttarakhand, Bihar, UP

PREPAREDNESS & SAFETY MEASURES

Before
- Follow building codes (earthquake-resistant structures).
- Secure heavy furniture, cylinders, electrical appliances.
- Be aware of evacuation routes.

During
- Drop – Cover – Hold (under a sturdy table).
- Stay away from windows, tall buildings, and electric poles.
- If outdoors, move to an open area.
- Do not use elevators.

After
- Check for injuries and gas/electric leaks.
- Be alert for aftershocks.
- Follow official instructions.
- Help others and stay calm.

KEY FACTS FOR EXAMS (SSC / UPSC)

- ✅ 90% of earthquakes occur in the Ring of Fire.
- ✅ Strongest earthquake ever: Valdivia, Chile (1960, M 9.5).
- ✅ Most earthquake-prone zone in India: Zone V.
- ✅ Earthquakes are measured by Richter Scale (now Mw) and their effects by Modified Mercalli Scale.
- ✅ P-waves are fastest; surface waves are most destructive.
- ✅ Induced earthquakes can be caused by large dams (e.g. Koyna, India).
- ✅ India lies in the Alpide Belt (Himalayan seismic zone).
- ✅ Earthquakes can also trigger tsunamis, landslides and volcanic eruptions.

QUICK ERVISION FLOW CHART

Tectonic Stress Builds Up
↓
Rocks Break Along a Fault (Release of Energy)
↓
Seismic Waves Travel (P, S and Surface Waves)
↓
Ground Shakes (Earthquake Felt on Surface)

REMEMBER !

- ✅ Earthquakes cannot be predicted accurately (yet).
- ✅ They can occur at any time.
- ✅ Preparedness saves lives.
- ✅ Invest in resilient infrastructure.
- ✅ Awareness and education reduce risks.

"When the Earth Shakes, Be Prepared, Not Scared."

A More Resilient World Is A Safer World !

KNOW | PREPARE | PRACTICE | STAY SAFE

Nature Reminds Us To Build A Safer Tomorrow !

Explore The Universe Beyond !

THE EXOSPHERE

EARTH'S OUTERMOST LAYER • GATEWAY TO SPACE

Same Sky Same Dreams Bigger You !

IMPORTANT FACTS ONLY • HIGHLY RELEVANT FOR SSC • EASY REVISION

WHAT IS EXOSPHERE?

- ☑ Outermost layer of Earth's atmosphere.
- ☑ Extends from ~700 km to 10,000 km (merges with outer space).
- ☑ Very thin air (mainly hydrogen & helium).
- ☑ Satellites orbit here (e.g. communication, weather, navigation).
- ☑ No definite upper limit. Gradually merges with space.

LAYERS OF ATMOSPHERE (Quick View)

Layer	Height (km)
Exosphere	700 – 10,0000
Thermosphere	80 – 700
Mesosphere	50 – 80
Stratosphere	12 – 50
Troposphere	0 – 12

Exosphere = Outermost layer (700 – 10,000 km)

KEY POINTS (EXOSPHERE)

- ☑ Least dense layer.
- ☑ Mainly Hydrogen & Helium.
- ☑ Temperature can be very high, but air is so thin that it feels cold.
- ☑ Satellites revolve in this layer.
- ☑ Merges with outer space.
- ☑ Important for: Communication, GPS, Weather forecasting.

Exosphere connects our Earth to the Universe !

THE SOLAR SYSTEM

8 Planets • 1 Sun • Countless Possibilities

PLANETS – IMPORTANT FACTS

Planet	Order from Sun	Special Points (SSC)
Mercury	1	Smallest planet, closest to Sun
Venus	2	Hottest planet
Earth	3	Only planet with life
Mars	4	Known as Red Planet
Jupiter	5	Largest planet
Saturn	6	Less dense than water (float)
Uranus	7	Coldest planet (tilted axis ~98°)
Neptune	8	Farthest planet

IMPORTANT SUPERLATIVES (PLANETS)

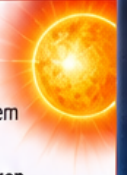
- 🏆 Largest planet → Jupiter
- 🏆 Smallest planet → Mercury
- 🏆 Hottest planet → Venus
- 🏆 Coldest planet → Uranus
- 🏆 Closest to Sun → Mercury
- 🏆 Farthest from Sun → Neptune
- 🏆 Fastest revolution (shortest year) → Mercury (88 days)
- 🏆 Slowest revolution (longest year) → Neptune (165 years)
- 🏆 Maximum number of moons → Saturn (as per latest count)
- 🏆 Minimum number of moons → Mercury & Venus (0)

MOONS – IMPORTANT FACTS

- ☑ Total natural moons (in Solar System) → 300+ (as per latest data)
- ☑ Maximum moons → Saturn
- ☑ Famous moons → Earth: 1 (Moon)
  Mars: 2 (Phobos, Deimos)
  Jupiter: 95+
  Saturn: 145+
  Uranus: 27+
  Neptune: 14+

- ☑ Largest moon (in Solar System) → Ganymede (Jupiter)
- ☑ Smallest moon (in Solar System) → S/2009 S 1 (Saturn) (≈ 0.3 km)
- ☑ Our Moon → Natural satellite of Earth

SUN – KEY FACTS



- ☑ A star (not a planet).
- ☑ Provides light and heat.
- ☑ Largest object in Solar System (~ 99% of total mass).
- ☑ Composed mainly of Hydrogen and Helium.
- ☑ Surface temperature → 5,500°C.
- ☑ Core temperature → 1.5 crore °C (15 million °C).
- ☑ Energy produced by nuclear fusion.

DAY, NIGHT, YEAR – IMPORTANT FACTS

Length of day (Earth)	→ 24 hours
Length of night (Earth)	→ 24 hours
Reason	→ Rotation of Earth
Length of year (Earth)	→ 365¼ days (≈ 365 days 6 hours)
Reason	→ Revolution around Sun

OTHER IMPORTANT FACTS

- Milky Way → Our galaxy.
- Andromeda → Nearest galaxy to Milky Way.
- Universe → All galaxies, stars, planets, etc.
- Biggest star → UY Scuti (largest known star yet).
- Smallest star → EBLM J0555-57Ab (one of this smallest known).

QUICK REVISION (ONE LINER)

- 📌 Exosphere → Outermost layer (700–10,000 km)
- 📌 Largest planet → Jupiter
- 📌 Smallest planet → Mercury
- 📌 Hottest planet → Venus
- 📌 Coldest planet → Uranus
- 📌 Closest to Sun → Mercury
- 📌 Farthest planet → Neptune
- 📌 Maximum moons → Saturn
- 📌 Minimum moons → Mercury & Venus (0)
- 📌 Largest moon → Ganymede (Jupiter)
- 📌 Smallest moon → S/2009 S 1 (Saturn)
- 📌 Length of day & night → 24 hours each
- 📌 Length of year → 365¼ days

TOP 10 LARGEST MOONS (BY SIZE)

Rank	Moon	Planet
1	Ganymede	Jupiter
2	Titan	Saturn
3	Callisto	Jupiter
4	Io	Jupiter
5	Moon (Earth)	Earth
6	Europa	Jupiter
7	Triton	Neptune
8	Titania	Uranus
9	Rhea	Saturn
10	Oberon	Uranus

MIND MAP – UNIVERSE AT A GLANCE

The more you explore, the more you realize how beautiful our planet is !

Small Planet Big Opportunities !

LEARN GEOGRAPHY | SCORE HIGHER | REACH FURTHER

Same Earth Brighter Future !

BIOSPHERE RESERVES OF INDIA

CONSERVING NATURE • SUPPORTING PEOPLE • SUSTAINING TOMORROW

Biodiversity
Ecosystems
Livelihoods
A Greener India !

People & Nature
Can Thrive Together !

Total Biosphere Reserves in India : 18
(As per latest updates)

"Biosphere Reserves are Living Laboratories for a Sustainable Future."

Diverse Regions
Unique Ecosystems
One Incredible India !

LIST OF BIOSPHERE RESERVES IN INDIA

No.	Biosphere Reserve	State/UT	Year of Notification	Key Features
1	Cold Desert	Himachal Pradesh	2009	High-altitude desert ecosystem
2	Nanda Devi	Uttarakhand	1988	Mountain ecosystem (UNESCO World Heritage Site)
3	Khangchendzonga	Sikkim	2000	Biodiversity-rich Himalayan region (UNESCO World Heritage Site)
4	Great Indian Desert	Rajasthan	1984	Thar Desert ecosystem
5	Rann of Kachchh	Gujarat	1986	Unique desert-wetland ecosystem
6	Gulf of Mannar	Tamil Nadu	1989	Marine biodiversity, coral reefs, seagrass beds
7	Panna	Madhya Pradesh	2011	Dry deciduous forests, ken river
8	Achanakmar-Amarkantak Maikal	Madhya Pradesh Chhattisgarh	2005	Central Indian highlands, source of Narmada, Son, Mahanadi
9	Similipal	Odisha	1994	Rich flora & fauna, tiger reserve
10	Pachmarhi	Madhya Pradesh	1999	Satpura range, diverse ecosystems
11	Dibru-Saikhowa	Assam	1997	Brahmaputra floodplains, rare species
12	Dihang-Dibang	Arunachal Pradesh	1998	Eastern Himalayas, rich biodiversity
13	Nokrek	Meghalaya	1988	Home of Red Panda (endemic species)
14	Manas	Assam	1989	Tiger reserve, elephant habitat (UNESCO World Heritage Site)
15	Sundarbans	West Bengal	1989	Mangrove ecosystem, Royal Bengal Tiger
16	Nilgiri	Tamil Nadu Kerala Karnataka	1986	Western Ghats, high endemism
17	Agasthyamalai	Kerala Tamil Nadu	2001	Rich in endemic species, Western Ghats
18	Great Nicobar (Indira Point)	Andaman & Nicobar Islands	2021	Tropical evergreen forests, unique island ecosystem

LEGEND

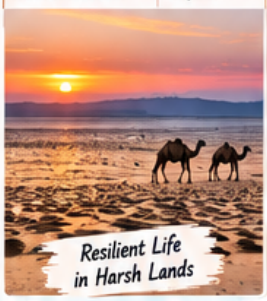
- Biosphere Reserve Location
- International Boundary
- State/UT Boundary
- Major River
- Himalayan Region
- Indo-Gangetic Plain
- Peninsular India
- Desert Region
- Coastal Region

HIMALAYAN BIOSPHERE RESERVES

Name	State/UT
Cold Desert	Himachal Pradesh
Nanda Devi	Uttarakhand
Khangchendzonga	Sikkim
Dihang-Dibang	Arunachal Pradesh
Nokrek	Meghalaya
Manas	Assam
Dibru-Saikhowa	Assam

DESERT / ARID REGION

Name	State/UT
Great Indian Desert	Rajasthan
Rann of Kachchh	Gujarat



CENTRAL INDIA

Name	State/UT
Panna	Madhya Pradesh
Achanakmar-Amarkantak Maikal	MP & Chhattisgarh
Similipal	Odisha
Pachmarhi	Madhya Pradesh

COASTAL & MARINE

Name	State/UT
Gulf of Mannar	Tamil Nadu
Sundarbans	West Bengal
Great Nicobar (Indira Point)	A&N Islands

WESTERN & SOUTHERN GHATS

Name	State/UT
Nilgiri	TN, Kerala, Karnataka
Agasthyamalai	Kerala & Tamil Nadu

KEY FACTS (FOR SSC CGL)

- India has 18 Biosphere Reserves (as per latest update).
- First Biosphere Reserve in India – Nilgiri (1986).
- Latest Biosphere Reserve – Great Nicobar (2021).
- Largest – Nanda Devi.
- Smallest – Nokrek.
- Only marine biosphere reserve – Gulf of Mannar.
- Sundarbans is a biosphere reserve, national park and Tiger Reserve.
- Both Nanda Devi and Manas are UNESCO World Heritage Sites.
- Biosphere reserves are part of UNESCO's Man and the Biosphere (MAB) Programme.

QUICK REVISION TRICKS

- ïigiri – First
- Great Nicobar – Latest
- Nanda Devi – Largest
- Nokrek – Smallest
- Gulf of Mannar – Only Marine
- Sundarbans – BR + NP + TR
- Manas & Nanda Devi – UNESCO
- Nokrek – Red Panda
- Achanakmar-Amarkantak – Source of Narmada, Son, Mahanadi

WHY BIOSPHERE RESERVES MATTER ?

- Conserve biodiversity and unique ecosystems.
- Support sustainable development.
- Protect traditional knowledge and local communities.
- Serve as natural laboratories for research and education.
- Help in climate change mitigation.
- Maintain ecological balance for future generations.

"Biosphere Reserves – Where Nature, People and Progress Coexist."

CONSERVE BIODIVERSITY | RESPECT NATURE | LEARN GEOGRAPHY | SCORE HIGHER

A Greener Today
A Better Tomorrow

NATIONAL PARKS OF INDIA

PROTECTING WILDLIFE | PRESERVING BIODIVERSITY | SECURING TOMORROW

More than 100 National Parks A Greener, Wilder Brighter India !

Different Landscapes One Incredible India !

NATIONAL PARKS – STATEWISE LIST
(Most Important for SSC CGL)

State/UT	Important National Parks
Jammu & Kashmir	Dachigam, Kishtwar
Ladakh	Hemis
Himachal Pradesh	Great Himalayan, Pin Valley
Uttarakhand	Jim Corbett, Rajaji, Valley of Flowers
Arunachal Pradesh	Namdapha, Mouling
Assam	Kaziranga, Manas, Orang
Meghalaya	Balpakram, Nokrek
Nagaland	Intanki
Manipur	Keibul Lamjao
Mizoram	Murlen
Tripura	Clouded Leopard
Sikkim	Khangchendzonga
West Bengal	Sundarbans, Jaldapara, Buxa, Neora Valley
Bihar	Valmiki
Jharkhand	Betla
Odisha	Similipal, Bhitarkanika
Chhattisgarh	Indravati, Kanger Valley
Madhya Pradesh	Kanha, Bandhavgarh, Pench (MP), Panna, Satpura, Sanjay-Dubri
Uttar Pradesh	Dudhwa, Pilibhit
Rajasthan	Ranthambore, Sariska, Desert, Keoladeo (Bharatpur)
Gujarat	Gir, Blackbuck (Velavadar), Marine (Gulf of Kachchh)
Maharashtra	Tadoba-Andhari, Navegaon, Sanjay Gandhi
Telangana	Kawal, Mrugavani
Andhra Pradesh	Sri Venkateswara
Karnataka	Bandipur, Nagarhole, Kudremukh
Kerala	Periyar, Silent Valley, Eravikulam
Tamil Nadu	Mukurthi, Guindy, Indira Gandhi (Wildlife)
Goa	Mollem (Bhagwan Mahavir)
Lakshadweep	Pitti (Marine)
Andaman & Nicobar	Mahatma Gandhi Marine
Dadra & Nagar Haveli & Daman & Diu	Dadra (Wildlife Sanctuary, near National Park)

MAP OF INDIA
SHOWING IMPORTANT NATIONAL PARKS

LEGEND

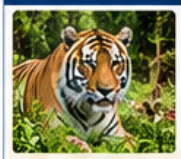
- Important National Park
- State Boundary
- International Boundary
- Water Body

National Parks are India's Natural Treasure Let's Protect Them !

FAMOUS NATIONAL PARKS & THEIR FLAGSHIP SPECIES

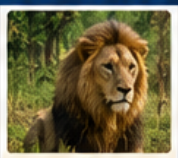
National Park	State	Flagship / Notable Species
Jim Corbett	Uttarakhand	Bengal Tiger, Asian Elephant
Kaziranga	Assam	Indian One-Horned Rhinoceros
Manas	Assam	Tiger, Elephant, Golden Langur
Sundarbans	West Bengal	Royal Bengal Tiger
Gir	Gujarat	Asiatic Lion
Ranthambore	Rajasthan	Bengal Tiger
Kanha	Madhya Pradesh	Barasingha (Swamp Deer)
Bandhavgarh	Madhya Pradesh	Bengal Tiger
Pench	MP/Maharashtra	Bengal Tiger (inspiration for The Jungle Book)
Similipal	Odisha	Elephant, Royal Bengal Tiger
Keoladeo (Bharatpur)	Rajasthan	Migratory Birds
Dachigam	J&K	Hangul (Kashmir Stag)
Hemis	Ladakh	Snow Leopard
Great Himalayan	Himachal Pradesh	Snow Leopard, Musk Deer
Valley of Flowers	Uttarakhand	Alpine Flora
Periyar	Kerala	Elephant, Tiger
Silent Valley	Kerala	Lion-tailed Macaque
Nagarhole	Karnataka	Tiger, Elephant
Bandipur	Karnataka	Tiger, Elephant
Kawal	Telangana	Tiger
Indravati	Chhattisgarh	Bison, Wild Dog
Desert	Rajasthan	Chinkara, Desert Fox
Marine (Gulf of Kachchh)	Gujarat	Marine Life, Corals

SOME ICONIC NATIONAL PARKS



Jim Corbett (Uttarakhand)
India's first national park (1936)

Kaziranga (Assam)
World's largest population of one-horned rhinoceros



Gir (Gujarat)
Only home of Asiatic Lion

Kanha (Madhya Pradesh)
Famous for Barasingha

Keoladeo (Bharatpur) (Rajasthan)
Paradise for migratory birds

Hemis (Ladakh)
Land of the Snow Leopard

KEY FACTS (SSC CGL)

- India has 106 National Parks (as of 2024).
- First National Park – Jim Corbett (1936).
- Largest National Park – Hemis (Ladakh).
- Smallest National Park – South Button (Andaman & Nicobar).
- Only home of Asiatic Lion – Gir (Gujarat).
- Highest diversity of organisms – Nanda Devi region (includes National Park & Biosphere Reserve).
- Kaziranga – famous for one-horned rhinoceros.
- Sundarbans – mangrove ecosystem & Royal Bengal Tiger.
- Keoladeo – UNESCO World Heritage Site.
- Valley of Flowers – UNESCO World Heritage Site.
- Use map-based questions to remember locations.
- Many national parks are part of larger Tiger Reserves.

QUICK REVISION TRICKS

- **Tiger** → Corbett, Ranthambore, Kanha, Bandhavgarh, Sundarbans
- **Rhino** → Kaziranga, Manas
- **Lion** → Gir (only place)
- **Elephant** → Periyar, Nagarhole, Bandipur
- **Barasingha** → Kanha
- **Birds** → Keoladeo (Bharatpur)
- **Snow Leopard** → Hemis, Great Himalayan
- **Flowers** → Valley of Flowers
- **Hangul** → Dachigam

DO YOU KNOW?

- Sundarbans is the world's largest mangrove forest.
- Keoladeo National Park is a man-made wetland.
- Hemis is the largest national park in India.
- South Button is the smallest national park in India.
- Gir is the only natural habitat of Asiatic Lion.
- Valley of Flowers is famous for alpine flowers and is in Uttarakhand.
- Pench inspired Rudyard Kipling's "The Jungle Book".
- Manas is a UNESCO World Heritage Site.
- Dachigam is famous for Hangul (Kashmir Stag).
- Silent Valley is known for Lion-tailed Macaque.

"Wildlife Today, A Safer Tomorrow"

GEOGRAPHY | SSC CGL | LEARN | CONSERVE | SUCCEED

MAJOR WATERFALLS OF THE WORLD

Location | Height | River | Country | Key Facts (SSC CGL)

Waterfalls Shape Landscapes, Power Economies, Inspire People !

Different Lands Different Falls Same Beauty One Amazing Planet !

NORTH AMERICA

No.	Waterfall	Height (m)	River	Country
1	Yosemite Falls	739	Merced River	USA
2	Niagara Falls	51	Niagara River	USA/Canada
3	Multnomah Falls	620	Multnomah Creek	USA
4	Palouse Falls	61	Palouse River	USA
5	Horseshoe Falls	57	Niagara River	Canada

SOUTH AMERICA

No.	Waterfall	Height (m)	River	Country
6	Angel Falls	979	Churún River	Venezuela
7	Iguazu Falls	82	Iguazu River	Argentina/Brazil
8	Gocta Falls	771	Cocahuayco River	Peru
9	Yumbilla Falls	896	Yumbilla River	Peru
10	Tres Hermanas Falls	914	Tambopata River	Peru

EUROPE

No.	Waterfall	Height (m)	River	Country
11	Rhine Falls	23	Rhine River	Switzerland
12	Dettifoss	44	Jökulsá á Fjöllum	Iceland
13	Kjostfossen	225	Kjos River	Norway
14	Krimml Falls	380	Krimml Ache	Austria
15	Duden Falls	40	Duden River	Turkey

ASIA

No.	Waterfall	Height (m)	River	Country
16	Jog Falls	253	Sharavathi River	India
17	Huangguoshu Falls	77	White Water River	China
18	Detian Falls	70	Quay Son River	China/Vietnam
19	Kegon Falls	97	Daiya River	Japan
20	Nohkalikai Falls	340	Kynshi River	India
21	Tad Fane Falls	120	Sé Pôn River	Laos

AFRICA

No.	Waterfall	Height (m)	River	Country
22	Victoria Falls	108	Zambezi River	Zambia/Zimbabwe
23	Tugela Falls	948	Tugela River	South Africa
24	Kalambo Falls	235	Kalambo River	Tanzania/Zambia
25	Epupa Falls	60	Kunene River	Namibia/Angola
26	Augrabies Falls	56	Orange River	South Africa

AUSTRALIA & OCEANIA

No.	Waterfall	Height (m)	River	Country
27	Sutherland Falls	580	Arthur River	New Zealand
28	Wallaman Falls	268	Stony Creek	Australia
29	Mackenzie Falls	109	MacKenzie River	Australia
30	Russell Falls	58	Russell River	Australia
31	Te Ana Falls	125	Te Ana River	New Zealand

KEY FACTS (SSC CGL)

- ✅ **Angel Falls** – World's highest waterfall (979 m)
- ✅ **Tugela Falls** – Highest waterfall in Africa (948 m)
- ✅ **Sutherland Falls** – Highest waterfall in Australia/Oceania (580 m)
- ✅ **Yosemite Falls** – Highest waterfall in North America (739 m)
- ✅ **Iguazu Falls** – Widest waterfall system in world
- ✅ **Victoria Falls** – One of the largest curtain waterfalls (UNESCO)
- ✅ **Jog Falls** – Highest waterfall in India (253 m)
- ✅ **Dettifoss** – Most powerful waterfall in Europe (by discharge)
- ✅ **Niagara Falls** – Famous for hydroelectric power & tourism

QUICK REVISION TIPS

- ✅ Remember highest waterfall of each continent.
- ✅ Note the river and country.
- ✅ Focus on superlatives (highest, widest, most powerful).
- ✅ Map-based questions are common in SSC CGL.
- ✅ Use mnemonics and visualize locations on the map.

"Waterfalls – Nature's Applause to the Earth."

GEOGRAPHY | SSC CGL | LEARN TODAY | EXPLORE TOMORROW

"Different Heights, Same Wonder – Our Beautiful Earth."

INDIA
PHYSICAL MAP
MOUNTAINS • PLATEAUS • PLAINS • DESERTS
RIVERS • COASTAL FEATURES • ISLANDS
INDIA'S EXTREME POINTS
Northernmost – Indira Col (Ladakh)
Southernmost – Indira Point (A&N Islands)
Easternmost – Kibithu (Arunachal Pradesh)
Westernmost – Guhar Moti (Gujarat)
IMPORTANT PHYSICAL FEATURES
MOUNTAIN RANGES
Himalayas (Karakoram, Ladakh, Zanskar, Pir Panjal, Dhauladhar, Shivalik)
Aravalli (oldest fold mountains)
Vindhya, Satpura
Western Ghats, Eastern Ghats
Nilgiri, Anamalai, Cardamom Hills
PLATEAUS
Deccan Plateau
Central Highlands (Malwa, Bundelkhand)
Chota Nagpur Plateau
Meghalaya Plateau
PLAINS
Indo-Gangetic Plain (Punjab, Haryana, UP, Bihar, WB)
Brahmaputra Plain (Assam)
DESERT
Thar Desert (Great Indian Desert)
RIVERS
Indus, Jhelum, Chenab, Ravi, Beas, Sutlej
Ganga, Yamuna, Brahmaputra
Narmada, Tapi
Godavari, Krishna, Cauvery
Mahanadi, Sabarmati, Mahi, Luni (others)
COASTS
Western Coast – Konkan, Malabar
Eastern Coast – Northern Circars, Coromandel
ISLANDS
Andaman & Nicobar (Bengal Bay)
Lakshadweep (Arabian Sea)
ALTITUDE (Height) COLOUR GUIDE
> 6,000 m
4,500 – 6,000 m
3,000 – 4,500 m
1,500 – 3,000 m
600 – 1,500 m
300 – 600 m
0 – 300 m
Sea Level
International Boundary
State Boundary
Peak (Height in metres)
0 250 500 750 1,000 km
N
W
E
S
K2 (8,611 m)
Karakoram Range
Ladakh Range
Zanskar Range
Pir Panjal
Dhauladhar
Shivalik
HIMALAYA
Kanchenjunga (8,586 m)
Indus
Jhelum
Chenab
Ravi
Sutlej
Ganga
Brahmaputra
Aravalli Range
Thar Desert
Indo-Gangetic Plain
Bundelkhand
Malwa Plateau
Vindhya Range
Narmada
Satpura Range
Tapi
Chota Nagpur Plateau
Naga Hills
Mizo Hills
Western Ghats
Eastern Ghats
Deccan Plateau
Godavari
Krishna
Cauvery
Konkan Coast
Malabar Coast
Coromandel Coast
Northern Circars
Nilgiri (2,637 m)
Anamalai
Cardamom Hills
Lakshadweep Islands (India)
Minicoy
Andaman & Nicobar Islands (India)
Indira Point
AFGHANISTAN
PAKISTAN
CHINA (TIBET)
NEPAL
BHUTAN
BANGLADESH
SRI LANKA
ARABIAN SEA
BAY OF BENGAL
INDIAN OCEAN
Know Your Land Know Your India !
Diverse Landforms Unique India !
Mountains to Oceans, Deserts to Forests, A Land of Diversity !
Study Maps Score Higher !
INDIA – A LAND OF CONTRASTS, A NATION OF POSSIBILITIES

LOCAL WINDS OF THE WORLD

Location | Direction | Type (Hot/Cold) | Key Facts (SSC CGL)

Winds Shape Places, People and Destinies !

Different Lands Different Winds Same Amazing Earth !

NORTH AMERICA

Wind	Location	Type	Key Fact
Chinook	Western Canada & USA	Warm	Raises temperature rapidly (snow melts)
Santa Ana	California (USA)	Warm	Hot, dry, strong wind (wildfires)
Norther	Texas & Gulf of Mexico	Cold	Cold wind from north, lowers temperature

SOUTH AMERICA

Wind	Location	Type	Key Fact
Pampero	Argentina & Uruguay	Cold	Cold, strong wind from south
Zonda	Andes (Argentina)	Warm	Hot, dry wind, common in winter/spring

EUROPE

Wind	Location	Type	Key Fact
Mistral	France (Rhône Valley)	Cold	Cold, dry wind from NW
Föhn	Alps (Central Europe)	Warm	Warm, dry wind (descending)

AFRICA

Wind	Location	Type	Key Fact
Harmattan	West Africa	Dry (Cold)	Dry, dusty wind from Sahara
Sirocco	North Africa (to Mediterranean)	Hot	Hot, dusty wind from Sahara

ASIA

Wind	Location	Type	Key Fact
Bora	Adriatic coast (N. Balkans)	Cold	Cold, gusty wind
Siberian	Siberia	Cold	Very cold, dry wind
Loo	North India (Indo-Gangetic Plains)	Hot	Hot, dry wind in summers
Karaburan	Central Asia (Kazakhstan, Uzbekistan)	Cold	Cold, dry wind
Blizzard	East Asia (China, Korea)	Cold	Severe cold wind with snow
Kamikaze	Japan	Cold	Strong, suicidal wind (stormy)

AUSTRALIA & OCEANIA

Wind	Location	Type	Key Fact
Willy-willy	Northern Australia	Warm	Hot, gusty wind (similar to dust devil)
Brickfielder	Interior Australia	Hot	Hot, dry wind
Southerly Buster	SE Australia	Cold	Cold wind from Southern Ocean
Nor'wester	New Zealand (South Island)	Warm	Warm, dry wind from northwest

IMPORTANT POINTS FOR SSC CGL

- ✓ **Chinook** → Snow eater (warm wind)
- ✓ **Santa Ana** → California (USA)
- ✓ **Pampero** → Argentina (cold)
- ✓ **Zonda** → Andes (hot & dry)
- ✓ **Mistral** → France (cold)
- ✓ **Föhn** → Warm wind in Alps (snow melts)
- ✓ **Harmattan** → West Africa (dusty)
- ✓ **Sirocco** → North Africa (hot & dusty)
- ✓ **Loo** → North India (hot summer wind)
- ✓ **Blizzard** → East Asia (severe cold)
- ✓ **Willy-willy** & **Brickfielder** → Australia (hot)
- ✓ **Nor'wester** → New Zealand (warm & dry)

TRICK TO REMEMBER

Hot Winds: Chinook, Santa Ana, Zonda, Föhn, Sirocco, Loo, Willy-willy, Brickfielder, Nor'wester

Cold Winds: Northeer, Pampero, Mistral, Bora, Siberian, Karaburan, Blizzard, Kamikaze, Southerly Buster

"Winds may be local, but their impact is global."

Small Winds Big Changes

A Small Part of a Vast Universe But a Big Part of a Beautiful Story !

THE UNIVERSE

EVERYTHING • EVERYWHERE • ALL AT ONCE

EXPLORE • UNDERSTAND • APPRECIATE • PROTECT

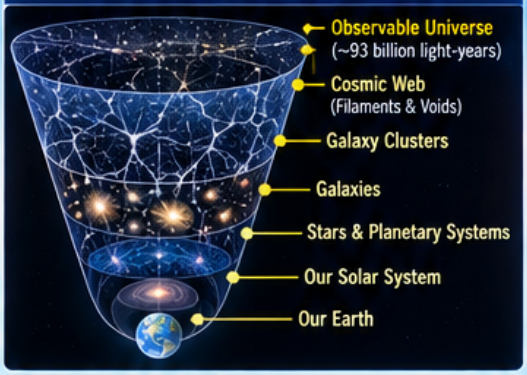
Same Origin
Same Space
Same Future
One Humanity !

WHAT IS THE UNIVERSE ?

- ✅ The Universe is the totality of everything that exists – space, time, matter, energy, galaxies, stars, planets and all cosmic structures.
- ✅ It is constantly expanding.
- ✅ It began with the Big Bang about 13.8 billion years ago.

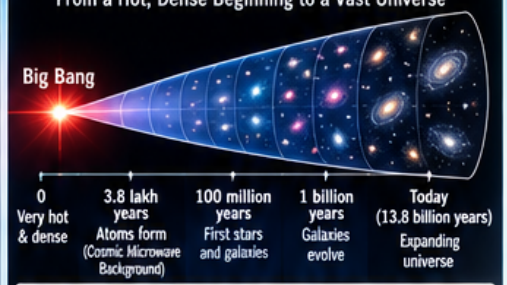
"The Universe is not just larger than we imagine, it is larger than we can imagine."

THE UNIVERSE AT A GLANCE



BIG BANG & EVOLUTION

From a Hot, Dense Beginning to a Vast Universe



The Universe is still expanding, and galaxies are moving away from each other (evidenced by Redshift).

SCALES OF THE UNIVERSE (SIZE COMPARISON)

Earth	Sun	Solar System	Milky Way Galaxy	Andromeda Galaxy	Observable Universe
~12,742 km	~1.39 million m km	~3 light-hours	~1,00,000 light-years	~2.5 million light-years	~93 billion light-years
Our Home	A Star	8 Planets	Our Galaxy	Our Neighbour	All We Can See (For Now)

From a tiny Earth to an almost infinite Universe !

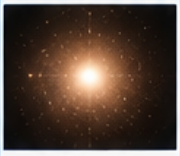
HIERARCHY OF COSMIC STRUCTURES

1. **Planet** (Earth)
2. **Star** (Sun)
3. **Solar System** (Sun + 8 planets + others)
4. **Galaxy** (Milky Way)
5. **Galaxy Group** (Local Group)
6. **Galaxy Cluster** (Virgo Cluster)
7. **Supercluster** (Laniakea Supercluster)
8. **Cosmic Web** (Filaments & Voids)
9. **Universe** (All of the above)

We are just a small part of a vast cosmic web of galaxies, clusters and superclusters.

"A tiny planet in a vast cosmic ocean."

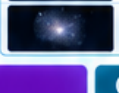
TYPES OF GALAXIES

Spiral Galaxy (e.g. Milky Way)	Elliptical Galaxy	Irregular Galaxy
		
• Disc-shaped • Spiral arms • Contains gas, dust, young stars	• Oval-shaped • Older stars • Less gas and dust	• No definite shape • Rich in gas and dust • Often formed by galactic collisions

OUR GALAXY – THE MILKY WAY

The Milky Way contains 100–400 billion stars.

OUR COSMIC NEIGHBOURS (LOCALLY IMPORTANT)

Andromeda Galaxy (M31)	• Nearest major galaxy ~2.5 million light-years • Will likely collide with Milky Way in ~4–5 billion years
Triangulum Galaxy (M33)	• Third largest in Local Group ~3 million light-years
Large Magellanic Cloud	• Dwarf galaxy ~163,000 light-years
 Small Magellanic Cloud	• Dwarf galaxy ~200,000 light-years

IMPORTANT STARS

Star	Key Fact
Sun	Nearest star to Earth
Proxima Centauri	Nearest star after Sun (~4.24 light-years)
Sirius	Brightest star in night sky
Canopus	Second brightest star
Arcturus	One of the brightest stars
Vega	Bright star in Lyra constellation
Betelgeuse	Red supergiant star
Rigel	Bright blue supergiant
Polaris (North Star)	Helps in navigation (in Ursa Minor)
UY Scuti	One of the largest known stars

NEBULA, STAR FORMATION & DEATH

Nebula	Star Formation	Death of a Star
• Cloud of gas and dust • Birthplace of stars • Example: Orion Nebula, Crab Nebula	• Starts in nebulae • Gas collapses due to gravity • Forms protostar → star	• Low mass → White dwarf (planetary nebula) • High mass → Supernova → Neutron star / Black hole • Example: Crab Nebula (supernova remnant)

OTHER IMPORTANT COSMIC OBJECTS

Black Hole	• Region where gravity is so strong that not even light can escape.
Neutron Star	• Extremely dense remnant of a supernova.
White Dwarf	• Remnant of a low-mass star.
Quasar	• Extremely bright active galaxy.
Pulsar	• Rapidly rotating neutron star.
Exoplanet	• Planet outside our solar system.
Dark Matter	• Invisible matter (doesn't emit light).
Dark Energy	• Mysterious energy causing accelerated expansion.
Cosmic Microwave Background (CMB)	• Radiatoun from early Universe (~3.8 lakh years after Big Bang).

KEY FACTS FOR SSC (QUICK REVISION)

- ✅ Universe began with the Big Bang (~13.8 billion years ago).
- ✅ It is still expanding (Redshift).
- ✅ Milky Way is a barred spiral galaxy.
- ✅ Nearest galaxy – Andromeda (~2.5 million light-years).
- ✅ Nearest star – Sun; nearest star after Sun – Proxima Centauri.
- ✅ Brightest star – Sirius.
- ✅ Largest known star – UY Scuti.
- ✅ Observable Universe – ~93 billion light-years in diameter.
- ✅ Light-year = unit of distance (9.46 trillion km).
- ✅ Earth is a tiny part of a vast cosmic web.

MIND MAP – UNIVERSE

SOME AMAZING NUMBERS

Age of Universe	13.8 billion years
Diameter of Observable Universe	~93 billion light-years
Age of Milky Way	~13.6 billion years
Number of galaxies (observable)	~2 trillion (estimated)
Number of stars in Milky Way	100–400 billion
Speed of light	3,00,000 km/s
1 light-year	9.46 trillion km
Time for sunlight to reach Earth	~8 minutes 20 seconds
Distance to Andromeda	~2.5 million light-years
Nearest star after Sun	Proxima Centauri (~4.24 ly)

Look Up
Learn More
Stay Curious !

SAME UNIVERSE | SAME ORIGIN | SAME FUTURE | A BRIGHTER HUMANITY

A Curiosity Today
A Better Tomorrow

One World Many Nations A Brighter You!

WORLD COUNTRIES – CAPITALS & CURRENCIES

FOR SSC CGL & ALL COMPETITIVE EXAMS

Maps Make Minds Stronger!

Abbreviations:
USD – US Dollar
EUR – Euro
INR – Indian Rupee
GBP – Pound Sterling
CNY – Chinese Yuan
JPY – Japanese Yen
AUD – Australian Dollar
CAD – Canadian Dollar
etc.

SSC CGL MUST REMEMBER (IMPORTANT TRAPS)

- **Australia** – Canberra – Australian Dollar
- **Turkey** – Ankara – Turkish Lira
- **Switzerland** – Bern – Swiss Franc
- **South Africa** – Pretoria (Executive) Cape Town (Legislative) Bloemfontein (Judicial) – Rand
- **Sri Lanka** – Sri Jayawardenepura Kotte – Sri Lankan Rupee
- **Kazakhstan** – Astana – Tenge
- **Myanmar** – Naypyidaw – Kyat
- **Côte d'Ivoire** – Yamoussoukro – CFA Franc
- **Tanzania** – Dodoma – Tanzanian Shilling
- **Eswatini** – Mbabane (Admin), Lobamba (Legislative) – Lilangeni
- **Netherlands** – Amsterdam – Euro
- **Denmark** – Copenhagen – Danish Krone
- **Norway** – Oslo – Norwegian Krone
- **Sweden** – Stockholm – Swedish Krona
- **Poland** – Warsaw – Zloty
- **Saudi Arabia** – Riyadh – Saudi Riyal
- **UAE** – Abu Dhabi – UAE Dirham

Study Smart ✓ Score High!

ASIA

Country	Capital	Currency
India	New Delhi	Indian Rupee (INR)
China	Beijing	Chinese Yuan (CNY)
Japan	Tokyo	Japanese Yen (JPY)
Pakistan	Islamabad	Pakistani Rupee (PKR)
Bangladesh	Dhaka	Bangladeshi Taka (BDT)
Sri Lanka	Sri Jayawardenepura Kotte	Sri Lankan Rupee (LKR)
Nepal	Kathmandu	Nepalese Rupee (NPR)
Bhutan	Thimphu	Ngultrum (BTN)
Myanarar	Naypyidaw	Kyat (MMK)
Thailand	Bangkok	Thai Baht (THB)
Malaysia	Kuala Lumpur	Malaysian Ringgit (MYR)
Singapore	Singapore	Singapore Dollar (SGD)
Indonesia	Jakarta	Indonesian Rupiah (IDR)
Philippines	Manila	Philippine Peso (PHP)
Vietnam	Hanoi	Vietnamese Dong (VND)
Turkey	Ankara	Turkish Lira (TRY)
Saudi Arabia	Riyadh	Saudi Riyal (SAR)
UAE	Abu Dhabi	UAE Dirham (AED)
Iran	Tehran	Iranian Rial (IRR)
Iraq	Baghdad	Iraqi Dinar (IQD)
Israel	Jerusalem	Israeli New Shekel (ILS)
Kazakhstan	Astana	Tenge (KZT)
Uzbekistan	Tashkent	Uzbek Som (UZS)

EUROPE

Country	Capital	Currency
United Kingdom	London	Pound Sterling (GBP)
Germany	Berlin	Euro (EUR)
France	Paris	Euro (EUR)
Italy	Rome	Euro (EUR)
Spain	Madrid	Euro (EUR)
Netherlands	Amsterdam	Euro (EUR)
Belgium	Brussels	Euro (EUR)
Switzerland	Bern	Swiss Franc (CHF)
Russia	Moscow	Russian Ruble (RUB)
Poland	Warsaw	Zloty (PLN)
Ukraine	Kyiv	Hryvnia (UAH)
Denmark	Copenhagen	Danish Krone (DKK)
Norway	Oslo	Norwegian Krone (NOK)
Sweden	Stockholm	Swedish Krona (SEK)
Finland	Helsinki	Euro (EUR)
Ireland	Dublin	Euro (EUR)
Portugal	Lisbon	Euro (EUR)
Greece	Athens	Euro (EUR)
Austria	Vienna	Euro (EUR)
Hungary	Budapest	Forint (HUF)
Czech Republic	Prague	Czech Koruna (CZK)
Romania	Bucharest	Romanian Leu (RON)
Bulgaria	Sofia	Bulgarian Lev (BGN)

AFRICA

Country	Capital	Currency
Algeria	Algiers	Algerian Dinar (DZD)
Egypt	Cairo	Egyptian Pound (EGP)
Nigeria	Abuja	Nigerian Naira (NGN)
South Africa	Pretoria	Rand (ZAR)
Kenya	Nairobi	Kenyan Shilling (KES)
Ethiopia	Addis Ababa	Ethiopian Birr (ETB)
Ghana	Accra	Ghanaian Cedi (GHS)
Morocco	Rabat	Moroccan Dirham (MAD)
Tunisia	Tunis	Tunisian Dinar (TND)
Tanzania	Dodoma	Tanzanian Shilling (TZS)
Uganda	Kampala	Ugandan Shilling (UGX)
Côte d'Ivoire	Yamoussoukro	CFA Franc (XOF)
Senegal	Dakar	CFA Franc (XOF)
DR Congo	Kinshasa	Congolese Franc (CDF)
Angola	Luanda	Angolan Kwanza (AOA)
Sudan	Khartoum	Sudanese Pound (SDG)
Zimbabwe	Harare	Zimbabwe Dollar (ZWL)
Cameroon	Yaoundé	CFA Franc (XAF)
Botswana	Gaborone	Pula (BWP)
Namibia	Windhoek	Namibian Dollar (NAD)
Eswatini	Mbabane / Lobamba	Lilangeni (SZL)
Lesotho	Maseru	Loti (LSL)
Madagascar	Antananarivo	Malagasy Ariary (MGA)

NORTH AMERICA

Country	Capital	Currency
United States	Washington D.C.	US Dollar (USD)
Canada	Ottawa	Canadian Dollar (CAD)
Mexico	Mexico City	Mexican Peso (MXN)
Guatemala	Guatemala City	Quetzal (GTQ)
Honduras	Tegucigalpa	Lempira (HNL)
El Salvador	San Salvador	US Dollar (USD)
Nicaragua	Managua	Córdoba (NIO)
Costa Rica	San José	Costa Rican Colón (CRC)
Panama	Panama City	Balboa (PAB)
Cuba	Havana	Cuban Peso (CUP)
Dominican Republic	Santo Domingo	Dominican Peso (DOP)
Haiti	Port-au-Prince	Gourde (HTG)
Jamaica	Kingston	Jamaican Dollar (JMD)
Trinidad & Tobago	Port of Spain	TT Dollar (TTD)

SOUTH AMERICA

Country	Capital	Currency
Brazil	Brasília	Brazilian Real (BRL)
Argentina	Buenos Aires	Argentine Peso (ARS)
Chile	Santiago	Chilean Peso (CLP)
Peru	Lima	Peruvian Sol (PEN)
Colombia	Bogotá	Colombian Peso (COP)
Venezuela	Caracas	Bolivar (VES)
Bolivia	Sucre	Boliviano (BOB)
Paraguay	Asunción	Guarani (PYG)
Uruguay	Montevideo	Uruguayan Peso (UYU)
Ecuador	Quito	US Dollar (USD)
Guyana	Georgetown	Guyanese Dollar (GYD)
Suriname	Paramaribo	Surinamese Dollar (SRD)

OCEANIA

Country	Capital	Currency
Australia	Canberra	Australian Dollar (AUD)
New Zealand	Wellington	New Zealand Dollar (NZD)
Fiji	Suva	Fijian Dollar (FJD)
Papua New Guinea	Port Moresby	Kina (PGK)
Samoa	Apia	Tala (WST)
Tonga	Nuku'alofa	Pa'anga (TOP)
Vanuatu	Port Vila	Vatu (VUV)
Marshall Islands	Majuro	US Dollar (USD)
Micronesia	Palikir	US Dollar (USD)
Solomon Islands	Honiara	Solomon Islands Dollar (SBD)
Kiribati	Tarawa	Australian Dollar (AUD)
Tuvalu	Funafuti	Australian Dollar (AUD)
Nauru	Yaren	Australian Dollar (AUD)
Palau	Ngerulmud	US Dollar (USD)

Explore Learn Grow

LEGEND ★ Capital City — Country Boundary — Ocean — Continent

"Different People | Different Currencies | One Shared Planet"

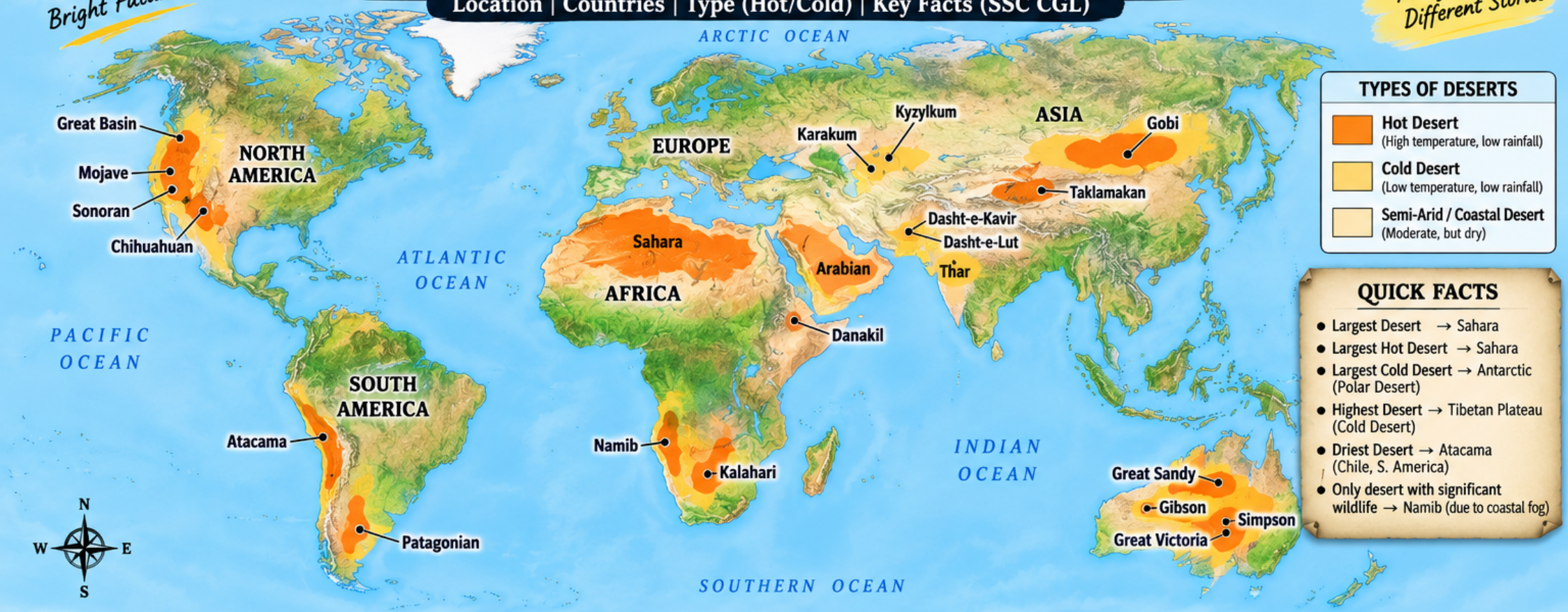
FACT: There are 195 internationally recognized countries in the world.

DESERTS OF THE WORLD

Location | Countries | Type (Hot/Cold) | Key Facts (SSC CGL)

Dry Lands Strong Facts Bright Future !

Same Planet Many Deserts Different Stories



TYPES OF DESERTS

- **Hot Desert** (High temperature, low rainfall)
- **Cold Desert** (Low temperature, low rainfall)
- **Semi-Arid / Coastal Desert** (Moderate, but dry)

QUICK FACTS

- Largest Desert → Sahara
- Largest Hot Desert → Sahara
- Largest Cold Desert → Antarctic (Polar Desert)
- Highest Desert → Tibetan Plateau (Cold Desert)
- Driest Desert → Atacama (Chile, S. America)
- Only desert with significant wildlife → Namib (due to coastal fog)

NORTH AMERICA

Desert	Location (Countries)	Type
Great Basin	USA	Cold
Mojave	USA	Hot
Sonoran	USA & Mexico	Hot
Chihuahuan	USA & Mexico	Hot

SOUTH AMERICA

Desert	Location (Countries)	Type
Atacama	Chile (Peru)	Coastal (Cold)
Patagonian	Argentina	Cold

AFRICA

Desert	Location (Countries)	Type
Sahara	North Africa (Many countries)	Hot
Namib	Namibia	Coastal (Cold)
Kalahari	Botswana, South Africa	Semi-Arid
Danakil	Ethiopia, Eritrea, Djibouti	Hot

ASIA

Desert	Location (Countries)	Type
Arabian	Saudi Arabia etc.	Hot
Thar	India & Pakistan	Hot
Dasht-e-Kavir	Iran	Cold
Dasht-e-Lut	Iran	Hot
Karakum	Turkmenistan	Cold
Kyzylkum	Uzbekistan, Kazakhstan	Cold
Taklamakan	China (Xinjiang)	Cold
Gobi	China & Mongolia	Cold

AUSTRALIA

Desert	Location (Countries)	Type
Great Sandy	Australia	Hot
Gibson	Australia	Hot
Great Victoria	Australia	Hot
Simpson	Australia	Hot

SSC CGL FOCUS POINTS

- Sahara → Largest & largest hot desert
- Antarctic → Largest cold desert → (polar desert)
- Atacama → Driest desert in world
- Namib → Coastal desert (fog), unique wildlife
- Thar → In India
- Dasht-e-Lut → One of the hottest places on Earth
- Gobi & Taklamakan → Cold deserts of Asia
- Patagonian → In Argentina (South America)

Deserts are not empty... They are full of extremes !

"Different Sands, Same Sky"

Rivers Shape Civilizations
Dams Build Progress
Lakes Sustain Life

INDIA'S WATER WEALTH

RIVERS • DAMS • LAKES — FLOWING THROUGH A BRIGHTER TOMORROW

Know
Remember
Revise
Excel !

IMPORTANT DAMS OF INDIA

Dam → River → State → Reservoir → Key Fact

Dam	River	State	Reservoir / Lake	Key Fact
Tehri Dam	Bhagirathi	Uttarakhand	Tehri Reservoir	One of the highest dams in India
Bhakra Nangal	Sutlej	HP/Punjab	Gobind Sagar	Major multipurpose project
Hirakud Dam	Mahanadi	Odisha	Hirakud Reservoir	One of the longest earthen dams
Sardar Sarovar	Narmada	Gujarat	Sardar Sarovar	Major Narmada Valley Project
Indira Sagar	Narmada	Madhya Pradesh	Indira Sagar	Large reservoir
Omkareshwar	Narmada	Madhya Pradesh	Omkareshwar	Narmada Valley Project
Nagarjuna Sagar	Krishna	Telangana/ AP	Nagarjuna Sagar	Major Krishna basin project
Srisailam	Krishna	AP/Telangana	Srisailam	Major hydroelectric project
Tungabhadra	Tungabhadra	Karnataka	Tungabhadra	Krishna basin
Almatti	Krishna	Karnataka	Almatti	Upper Krishna Project
Koyna	Koyna	Maharashtra	Shivsagar	Major hydroelectric project
Jayakwadi	Godavari	Maharashtra	Nath Sagar	Important irrigation project
Mettur	Kaveri	Tamil Nadu	Stanley	Major Kaveri project
Krishnarajasagara (KRS)	Kaveri	Karnataka	Krishna Raja Sagara	Near Mysuru
Idukki	Periyar	Kerala	Idukki	Famous arch dam
Mullaperiyar	Periyar	Kerala	Mullaperiyar	Inter-state water issue
Rihand	Rihand	Uttar Pradesh	Govind Ballabh Pant Sagar	Major reservoir of UP
Gandhi Sagar	Chambal	Madhya Pradesh	Gandhi Sagar	Chambal Valley Project
Rana Pratap Sagar	Chambal	Rajasthan	Rana Pratap Sagar	Chambal Project
Kota Barrage	Chambal	Rajasthan	–	Irrigation structure
Ukai	Tapi	Gujarat	Ukai	Major Tapi basin project
Maithon	Barakar	Jharkhand	Maithon	DVC project
Panchet	Damodar	Jharkhand	Panchet	DVC project
Tilaiya	Barakar	Jharkhand	Tilaiya	First DVC dam
Konar	Konar	Jharkhand	Konar	DVC project

IMPORTANT RIVERS OF INDIA

River → Source → States → Flows Into → Key Fact

River	Source	Major States	Flows Into	Key Fact
Ganga	Gangotri Glacier	UK, UP, Bihar, WB	Bay of Bengal	National river
Yamuna	Yamunotri Glacier	UK, Haryana, Delhi, UP	Ganga	Major tributary
Brahmaputra	Tibet (Tsangpo)	Arunachal, Assam	Bay of Bengal	Called Tsangpo in Tibet
Indus	Tibet	Ladakh	Arabian Sea	Major Himalayan river
Sutlej	Tibet	HP, Punjab	Indus	Bhakra Nangal
Beas	HP	HP, Punjab	Sutlej	Only Indian tributary
Ravi	HP	HP, Punjab	Chenab	Indus system
Chenab	HP	HP, J&K	Indus	Formed by Chandra + Bhaga
Jhelum	Kashmir	J&K	Chenab	Flows through Wular Lake
Narmada	Amarkantak	MP, Maharashtra, Gujarat	Arabian Sea	West-flowing
Tapi	Multai (MP)	MP, Maharashtra, Gujarat	Arabian Sea	West-flowing
Godavari	Trimbakeshwar	Maharashtra, Telangana, AP	Bay of Bengal	Dakshin Ganga
Krishna	Mahabaleshwar	Maharashtra, Karnataka, Telangana, AP	Bay of Bengal	Peninsular river
Mahanadi	Chhattisgarh	Chhattisgarh, Odisha	Bay of Bengal	Hirakud
Kaveri	Talakaveri	Karnataka, Tamil Nadu	Bay of Bengal	Important Indian river
Sabarmati	Aravalli	Rajasthan, Gujarat	Gulf of Khambhat	West-flowing
Luni	Aravalli	Rajasthan	Rann of Kachchh	Inland drainage
Mahi	MP	MP, Rajasthan, Gujarat	Gulf of Khambhat	Crosses Tropic of Cancer
Periyar	Western Ghats	Kerala	Arabian Sea	Longest river of Kerala

IMPORTANT LAKES OF INDIA

Lake → State/UT → Type → Key Fact

Lake	State/UT	Type	Key Fact
Wular	J&K	Freshwater	One of the largest freshwater lakes
Dal	J&K	Freshwater	Famous for houseboats
Pangong Tso	Ladakh	Brackish	Transboundary (India–China)
Tso Moriri	Ladakh	High-altitude	Ramsar site
Chilika	Odisha	Brackish	Largest coastal lagoon in India
Pulicat	Andhra Pradesh / Tamil Nadu	Brackish	Second largest brackish lagoon
Kolleru	Andhra Pradesh	Freshwater	Between Krishna & Godavari deltas
Loktak	Manipur	Freshwater	Famous for Phumdis
Sambhar	Rajasthan	Saline	Largest inland salt lake
Vembanad	Kerala	Brackish	Longest lake in India
Ashtamudi	Kerala	Backwater	Ramsar site
Sasthamkotta	Kerala	Freshwater	Largest freshwater lake in Kerala
Nainital	Uttarakhand	Freshwater	Kumaon Himalayas
Bhojtal (Upper Lake)	Madhya Pradesh	Freshwater	Bhopal
Lonar	Maharashtra	Crater	Formed by meteorite impact
Hussain Sagar	Telangana	Artificial	Hyderabad
Nakki	Rajasthan	Freshwater	Mount Abu
Umiam	Meghalaya	Artificial	Near Shillong

MAJOR DAM SYSTEMS (FLOW CHART)

- Sutlej → Bhakra Nangal → Gobind Sagar
- Bhagirathi → Tehri → Tehri Reservoir
- Mahanadi → Hirakud → Hirakud Reservoir
- Narmada → Indira Sagar → Omkareshwar → Sardar Sarovar
- Krishna → Almatti → Srisailam → Nagarjuna Sagar
- Kaveri → KRS → Mettur → Stanley Reservoir
- Chambal → Gandhi Sagar → Rana Pratap Sagar → Kota Barrage
- Damodar–Barakar → Tilaiya → Konar → Maithon → Panchet

MAJOR RIVER SYSTEMS (FLOW CHART)

INDUS SYSTEM
Indus
→ Jhelum
→ Chenab
→ Ravi
→ Beas
→ Sutlej

GANGA SYSTEM
Ganga
→ Yamuna (with tributaries)

PENINSULAR RIVERS
- Narmada (West)
- Tapi (West)
- Godavari (East)
- Krishna (East)
- Mahanadi (East)
- Kaveri (East)

OTHER IMPORTANT RIVERS

- Sabarmati → Gulf of Khambhat
- Mahi → Gulf of Khambhat
- Luni → Rann of Kachchh
- Periyar → Arabian Sea

SPECIAL LAKES – QUICK FLOW CHART

- Largest Inland Salt Lake → Sambhar (Rajasthan)
- Largest Coastal Lagoon → Chilika (Odisha)
- Largest Freshwater Lake → Wular (J&K)
- Crater Lake → Lonar (Maharashtra)
- Famous for Phumdis → Loktak (Manipur)
- Houseboats → Dal Lake (Srinagar)
- Between Krishna & Godavari Deltas → Kolleru (Andhra Pradesh)
- Second Largest Brackish Lagoon → Pulicat (AP & TN)
- Longest Lake in India → Vembanad (Kerala)

KEY FACTS & TRICKS (SSC CGL)

- ✅ Tehri Dam – on Bhagirathi (not Alaknanda)
- ✅ Hirakud – on Mahanadi
- ✅ Sardar Sarovar – on Narmada
- ✅ Mettur – on Kaveri
- ✅ Bhakra – on Sutlej
- ✅ Idukki – on Periyar
- ✅ Chilika – largest coastal lagoon
- ✅ Sambhar – largest inland salt lake
- ✅ Lonar – only crater lake
- ✅ Loktak – famous for phumdis
- ✅ Kolleru – between Krishna & Godavari
- ✅ Vembanad – longest lake
- ✅ Pangong Tso – India-China (Ladakh)
- ✅ Jhelum – flows through Wular Lake

Rivers Give Life
Dams Give Power
Lakes Give Balance
Together They Build a Stronger India

WHY IMPORTANT FOR SSC CGL?

- ✅ Direct and pair-based questions are common.
- ✅ Asked in previous years (Dams–Rivers–Lakes).
- ✅ Important for Static GK & Geography section.
- ✅ Also useful for State PCS, UPSC, Railways, Defence exams.

"Water is not just a resource, it is our responsibility."

CONSERVE WATER | KNOW YOUR GEOGRAPHY | SCORE HIGHER

OCEAN CURRENTS OF THE WORLD

Location | Type (Warm/Cold) | Key Facts (SSC CGL)

Ocean Currents Move Heat, Shape Climate !

Same Ocean Different Currents Different Climate !

Types of Ocean Currents

- **Warm Current** (Brings warm water, increases temperature & rainfall)
- **Cold Current** (Brings cold water, decreases temperature & rainfall)

Key Facts

- **Gulf Stream** – moderates climate of Western Europe.
- **Kuroshio** – warm current, Japan.
- **Humboldt (Peru)** – rich fishing (upwelling).
- **Benguela** – rich fishing (upwelling).
- **Antarctic Circumpolar** – only current that flows around Antarctica.
- **Agulhas** – warm current, Indian Ocean.
- **Labrador** – cold current, North Atlantic.

Remember
Warm = Red
Cold = Blue

MAJOR OCEAN CURRENTS (SSC CGL – IMPORTANT)

Atlantic Ocean

Current	Type	Location / Key Fact
Gulf Stream	Warm	N. Atlantic (eastern US to Europe)
North Atlantic Current	Warm	Europe's west coast (moderates climate)
Canary Current	Cold	NW Africa (desert region)
Benguela Current	Cold	SW Africa (rich fishing)
Brazil Current	Warm	E. coast of S. America
Falkland Current	Cold	S. Atlantic (fishing)
Labrador Current	Cold	E. Canada (ice, cold climate)

Pacific Ocean

Current	Type	Location / Key Fact
Kuroshio Current	Warm	Japan (warm, humid)
Oyashio Current	Cold	N. Pacific (rich fishing)
California Current	Cold	W. coast of N. America (fishing)
Alaska Current	Cold	Alaska (cold, rich fishing)
North Equatorial Current	Warm	Tropics (east → west)
South Equatorial Current	Cold	Tropics (east → west)
Peru (Humboldt) Current	Cold	S. America (rich fishing)

Indian Ocean

Current	Type	Location / Key Fact
Agulhas Current	Warm	E. Africa (warm, humid)
West Australian Current	Cold	W. Australia (cool, dry)
Monsoon Current	Seasonal	Reverses with monsoon
Equatorial Current	Warm	Near equator (east → west)

Southern Ocean

Current	Type	Location / Key Fact
Antarctic Circumpolar	Cold	Surrounds Antarctica (only current)

Exam Focus / Must Remember

- ✓ **Gulf Stream** → Europe (warm)
- ✓ **Kuroshio** → Japan (warm)
- ✓ **Humboldt** → Peru (cold, fishing)
- ✓ **Benguela** → SW Africa (cold, fishing)
- ✓ **Agulhas** → E. Africa (warm)
- ✓ **Labrador** → E. Canada (cold)
- ✓ **Antarctic Circumpolar** → around Antarctica
- ✓ **Peru + Benguela** → rich fishing (upwelling)

Quick Trick

Warm = Toward Poles (generally)
Cold = Toward Equator (generally)

Ocean Currents = Climate | Climate = Life

IMPORTANT SEAS OF THE WORLD

Location | Connected Oceans | Nearby Countries | Key Facts (SSC CGL)

Seas connect lands, shape climates, drive trade and tell stories of civilizations !

Different Seas Different Lands One Connected World !

- Marginal Sea (edge of ocean)
- Mediterranean Sea (enclosed by land)
- Inland Sea (landlocked)

KEY FACTS

- ✓ Seas are smaller than oceans.
- ✓ Marginal seas lie along the edge of oceans.
- ✓ Mediterranean seas are almost enclosed by land.
- ✓ Inland seas have no direct connection to oceans.
- ✓ Important for trade, fisheries, oil & gas and climate.
- ✓ Frequently asked in SSC CGL and other exams.

Learn the Seas, Explore the World !

SEAS OF THE ATLANTIC OCEAN

Sea	Location / Between	Type
Caribbean Sea	N. & S. America	Marginal
Sargasso Sea	Central Atlantic	Marginal
Gulf of Mexico	N. America	Marginal
North Sea	Europe	Marginal
Baltic Sea	N. Europe	Marginal
Norwegian Sea	Europe	Marginal
Celtic Sea	Europe	Marginal
English Channel	UK – France	Marginal

SEAS OF THE PACIFIC OCEAN

Sea	Location / Between	Type
Bering Sea	USA – Russia	Marginal
Sea of Okhotsk	Russia – Japan	Marginal
Sea of Japan (East Sea)	Japan – Korea – Russia	Marginal
Yellow Sea	China – Korea	Marginal
East China Sea	China – Japan	Marginal
South China Sea	China – SE Asia	Marginal
Philippine Sea	Philippines – Pacific	Marginal
Coral Sea	Australia – Pacific	Marginal
Tasman Sea	Australia – New Zealand	Marginal

SEAS OF THE INDIAN OCEAN

Sea	Location / Between	Type
Arabian Sea	India – Arabia	Marginal
Bay of Bengal	India – SE Asia	Marginal
Andaman Sea	India – Myanmar – Thailand	Marginal
Laccadive Sea	India – Maldives	Marginal
Gulf of Aden	Arabia – Africa	Marginal
Red Sea	Africa – Arabia	Mediterranean
Java Sea	Indonesia	Marginal
Banda Sea	Indonesia	Marginal
Arafura Sea	Australia – Indonesia	Marginal
Timor Sea	Australia – Timor	Marginal

SEAS OF THE ARCTIC OCEAN

Sea	Location	Type
Barents Sea	N. Europe – Russia	Marginal
Kara Sea	Russia	Marginal
Laptev Sea	Russia	Marginal
East Siberian Sea	Russia	Marginal
Chukchi Sea	Russia – USA	Marginal
Beaufort Sea	USA – Canada	Marginal
White Sea	Russia	Marginal

INLAND SEAS

Sea	Location	Note
Caspian Sea	Between Europe & Asia	Largest inland sea
Sea of Azov	Ukraine – Russia	Connected to Black Sea (narrow)

OTHER IMPORTANT SEAS

Sea	Location / Between	Type
Mediterranean Sea	Europe – Africa – Asia	Mediterranean
Black Sea	Europe – Asia	Mediterranean
Weddell Sea	Near Antarctica (Atlantic sector)	Marginal
Ross Sea	Near Antarctica (Pacific sector)	Marginal

TRICKS TO REMEMBER (SSC CGL)

- ✓ **Red Sea** → Red colour (important trade route).
- ✓ **Black Sea** → Black (name easy).
- ✓ **Caspian Sea** → Largest inland sea.
- ✓ **Baltic** → Between Baltic countries (N. Europe).
- ✓ **Bering Sea** → Between two 'B's (USA & Russia).
- ✓ **Bay of Bengal** → Near Bengal (India).

"Different Seas, Same Planet, One Blue Home."

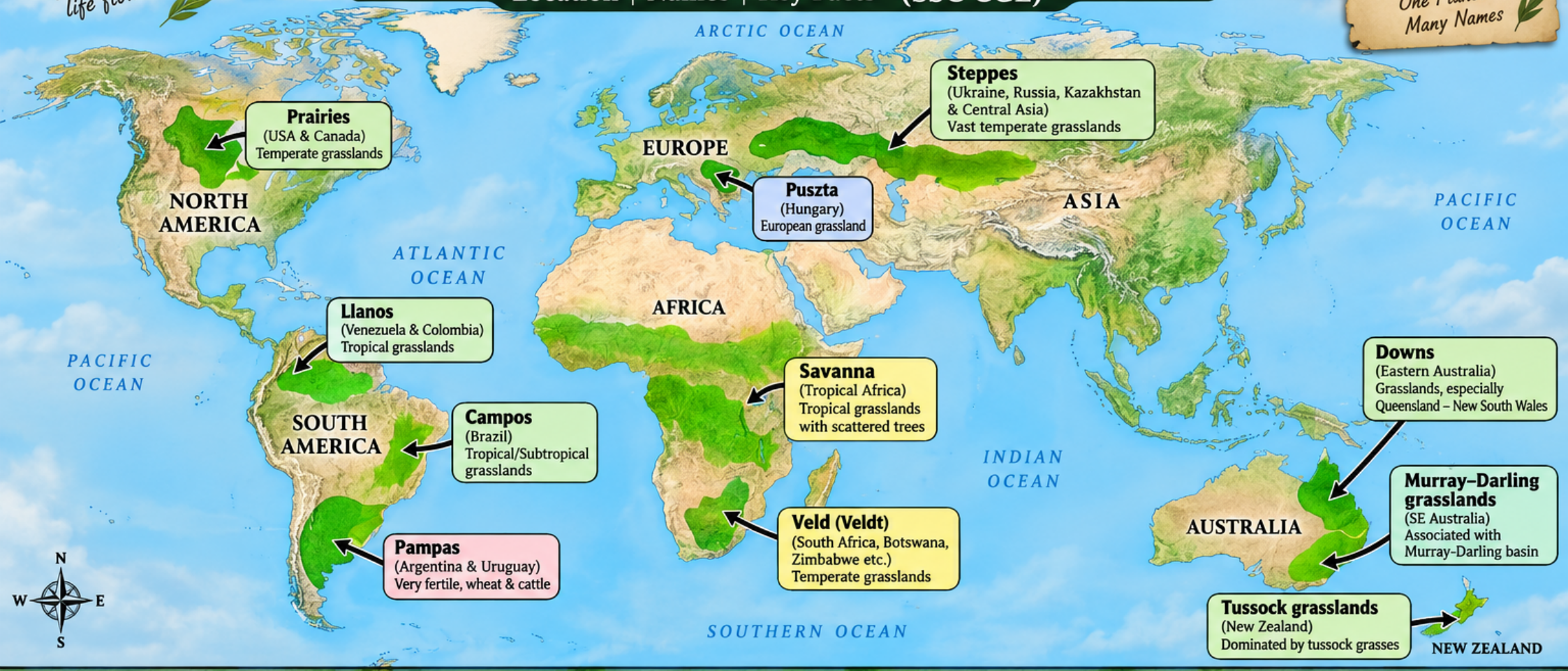
GEOGRAPHY | SSC CGL | MAP IT | LEARN IT | REMEMBER IT

GRASSLANDS OF THE WORLD

Location | Names | Key Facts (SSC CGL)

Where the grass grows, life flows ...

Different Lands Same Grass One Planet Many Names



QUICK REFERENCE TABLE

Grassland	Location / Countries	Type / Key Fact
Prairies	USA & Canada	Temperate grasslands
Pampas	Argentina & Uruguay	Very fertile; wheat & cattle
Llanos	Venezuela & Colombia	Tropical grasslands
Campos	Brazil	Tropical/Subtropical grasslands
Savanna	Tropical Africa	Tropical grassland with scattered trees
Veld (Veldt)	South Africa, Botswana, Zimbabwe etc.	Temperate grasslands
Steppes	Ukraine, Russia, Kazakhstan & Central Asia	Vast temperate grasslands
Puszta	Hungary	European grassland
Downs	Eastern Australia	Grasslands, esp. Queensland–New South Wales
Murray–Darling	SE Australia	Associated with Murray-Darling basin
Tussock grasslands	New Zealand	Dominated by tussock grasses

TOP 7 FOR SSC CGL

1. **Prairies** → North America
2. **Pampas** → Argentina & Uruguay
3. **Llanos** → Venezuela & Colombia
4. **Campos** → Brazil
5. **Veld** → South Africa
6. **Steppes** → Central Asia
7. **Downs** → Australia

MEMORY TRICK

P-P-L-C-V-S-D

- **P** Prairies → North America
- **P** Pampas – Argentina
- **L** Llanos – Venezuela
- **C** Campos – Brazil
- **V** Veld – South Africa
- **S** Steppes – Central Asia
- **D** Downs → Australia

IMPORTANT FACTS

Exam Focus ✓

- ✓ **Prairies, Steppes, Veld** → Temperate grasslands
- ✓ **Savanna, Llanos, Campos** → Tropical grasslands
- ✓ **Pampas** → Very fertile (wheat & cattle)
- ✓ **Puszta** → Hungary (Europe)
- ✓ **Downs** → Eastern Australia
- ✓ **Tussock grasslands** → New Zealand
- ✓ **Savanna** is the largest tropical grassland in the world.
- ✓ **Pampas** is one of the most fertile grasslands.
- ✓ **Steppes** are the largest temperate grasslands.
- ✓ **Veld** is the grassland of South Africa.

Grasslands Feed the World

GEOGRAPHY | WORLD | SSC CGL | LEARN SMART | SCORE HIGH

IMPORTANT BAYS & GULFS OF THE WORLD

Location | Surrounding Countries | Connected Seas/Oceans | Key Facts (SSC CGL)

Bays & Gulfs Shape Coasts, Connect Continents, Create Opportunities !

Small Water Bodies Big Role in Trade, History & Geography !

Explore the World Through Its Coasts !

IMPORTANT BAYS OF THE WORLD

Bay	Surrounded By (Countries/Regions)	Connected To
Bay of Bengal	India, Bangladesh, Myanmar, Sri Lanka	Indian Ocean
Hudson Bay	Canada	Atlantic Ocean
Bay of Biscay	France, Spain	Atlantic Ocean
Chesapeake Bay	USA	Atlantic Ocean
Tokyo Bay	Japan	Pacific Ocean
Osaka Bay	Japan	Pacific Ocean
Manila Bay	Philippines	Pacific Ocean
Gulf of Guayaquil (Bay)	Ecuador, Peru	Pacific Ocean
La Plata (Rio de la Plata)	Argentina, Uruguay	Atlantic Ocean
Great Australian Bight	Australia	Southern Ocean

IMPORTANT GULFS OF THE WORLD

Gulf	Surrounded By (Countries/Regions)	Connected To
Persian Gulf	Iran, Iraq, Kuwait, Saudi Arabia, Bahrain, Qatar, UAE	Gulf of Oman (Indian Ocean)
Gulf of Mexico	USA, Mexico, Cuba	Atlantic Ocean
Gulf of Alaska	USA (Alaska)	Pacific Ocean
Gulf of California	Mexico	Pacific Ocean
Gulf of St. Lawrence	Canada	Atlantic Ocean
Gulf of Guinea	West African Countries	Atlantic Ocean
Gulf of Aden	Yemen, Somalia, Djibouti	Indian Ocean
Gulf of Oman	Iran, Pakistan, Oman	Arabian Sea
Gulf of Aqaba	Egypt, Israel, Jordan, Saudi Arabia	Red Sea
Gulf of Suez	Egypt	Red Sea

MORE IMPORTANT GULFS

Gulf	Surrounded By	Connected To
Gulf of Kachchh	India (Gujarat)	Arabian Sea
Gulf of Khambhat	India (Gujarat)	Arabian Sea
Gulf of Mannar	India, Sri Lanka	Palk Bay (Indian Ocean)
Gulf of Thailand	Thailand, Cambodia, Vietnam, Malaysia	South China Sea
Gulf of Tonkin	Vietnam, China	South China Sea
Gulf of Martaban	Myanmar	Andaman Sea
Gulf of Carpentaria	Australia	Arafura Sea
Bohai Gulf	China	Yellow Sea
Korea Bay	North & South Korea, China	Yellow Sea
Gulf of Bothnia	Sweden, Finland	Baltic Sea
Gulf of Finland	Finland, Estonia, Russia	Baltic Sea
Gulf of Riga	Latvia, Estonia	Baltic Sea
Gulf of Gabes	Tunisia, Libya	Mediterranean Sea
Gulf of Sirte	Libya	Mediterranean Sea

OTHER NOTABLE BAYS & GULFS

Bay / Gulf	Surrounded By	Connected To
Palk Bay	India, Sri Lanka	Gulf of Mannar (Indian Ocean)
Saronic Gulf	Greece	Aegean Sea
Gulf of Lions	France	Mediterranean Sea
Gulf of Sidra	Libya	Mediterranean Sea
Gulf of Honduras	Honduras, Nicaragua, Belize	Caribbean Sea
Gulf of Paria	Venezuela, Trinidad & Tobago	Atlantic Ocean
Hangzhou Bay	China	East China Sea
Sulu Bay	Philippines	Sulu Sea
Cam Ranh Bay	Vietnam	South China Sea
Subic Bay	Philippines	South China Sea
Gulf of Tartary	Russia	Sea of Japan
Gulf of Lûte	Iran	Persian Gulf
Gulf of Penas	Chile	Pacific Ocean
Gulf of San Jorge	Argentina	Atlantic Ocean

KEY FACTS & TRICKS (SSC CGL)

- **Bay** – larger, partially enclosed by land.
- **Gulf** – smaller than a bay, more enclosed.
- **Gulf of Mexico** – connects to Atlantic Ocean (important for trade & oil).
- **Persian Gulf** – world's most important oil region.
- **Bay of Bengal** – largest bay in the world.
- **Hudson Bay** – almost landlocked, cold region.
- **Palk Bay & Gulf of Mannar** – between India & Sri Lanka.
- **Gulf of Kachchh & Gulf of Khambhat** – on the west coast of India.
- **Gulf of Thailand** – part of South China Sea.
- **Great Australian Bight** – open bay on south coast of Australia.
- **Bohai Gulf** – important for China's economy.
- **Remember:** Bays & Gulfs are important for trade, ports, fisheries, climate and strategic military locations.

"From the Deepest Bays to the Busiest Gulfs – The World is More Connected Than We Think."

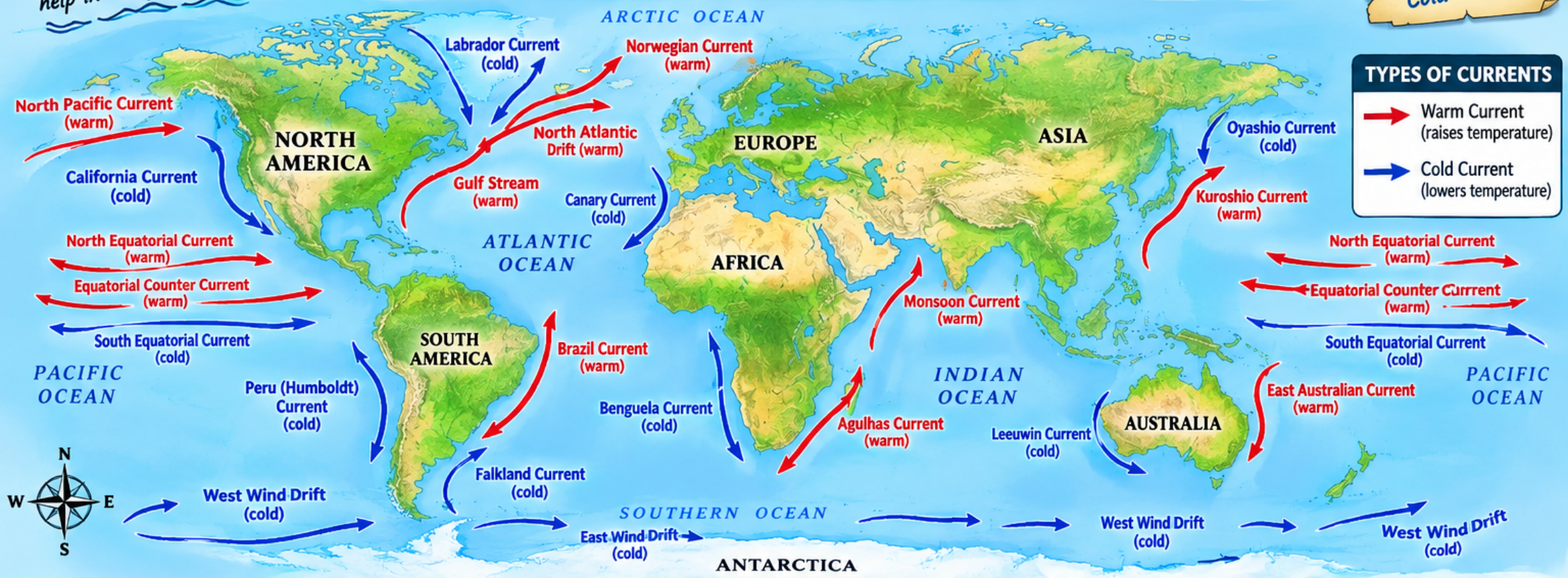
GEOGRAPHY | SSC CGL | LEARN | PRACTICE | SUCCEED

OCEAN CURRENTS OF THE WORLD

Location | Type (Warm/Cold) | Key Facts (SSC CGL)

Ocean currents regulate climate, bring nutrients & help in navigation!

Remember:
Warm = Red
Cold = Blue



TYPES OF CURRENTS

- Warm Current (raises temperature)
- Cold Current (lowers temperature)

MAJOR WARM CURRENTS

1. **Gulf Stream** – N. Atlantic (USA → Europe)
2. **North Atlantic Drift** – N. Atlantic
3. **Kuroshio Current** – W. Pacific (Japan)
4. **Brazil Current** – E. S. America
5. **Agulhas Current** – Indian Ocean (S. Africa)
6. **East Australian Current** – E. Australia
7. **North Pacific Current** – N. Pacific
8. **Equatorial Currents** – all oceans (except S. polar)
9. **Monsoon Current** – Indian Ocean (seasonal)

MAJOR COLD CURRENTS

1. **Labrador Current** – N. Atlantic (Canada)
2. **California Current** – NE. Pacific (USA)
3. **Peru (Humboldt) Current** – E. Pacific (Chile, Peru)
4. **Canary Current** – NE. Atlantic (NW Africa)
5. **Benguela Current** – SW. Africa
6. **Oyashio Current** – N. Pacific (Japan)
7. **Falkland Current** – S. Atlantic (Argentina)
8. **West Australian Current** – SW. Australia
9. **East Greenland Current** – Arctic region
10. **Antarctic Circumpolar Current** – around Antarctica

IMPORTANT FACTS

- ✓ Warm currents flow from equator to poles (usually).
- ✓ Cold currents flow from poles to equator (usually).
- ✓ **Gulf Stream** keeps Europe warmer.
- ✓ Peru current brings rich nutrients → good fishing.
- ✓ **El Niño & La Niña** are due to changes in equatorial Pacific currents.
- ✓ Currents affect climate, rainfall, marine life and navigation.

QUICK TRICKS

G (Gulf Stream) – N. Atlantic (warm)
K (Kuroshio) → W. Pacific (warm)
B (Brazil) → E. S. America (warm)
P (Peru) → W. S. America (cold)
C (California) – E. Pacific (cold)
L (Labrador) → N. Atlantic (cold)
B (Benguela) → SW. Africa (cold)
A (Agulhas) → SE. Africa (warm)

"Currents connect the oceans, and oceans connect the world."

IMPORTANT EXAM POINTS

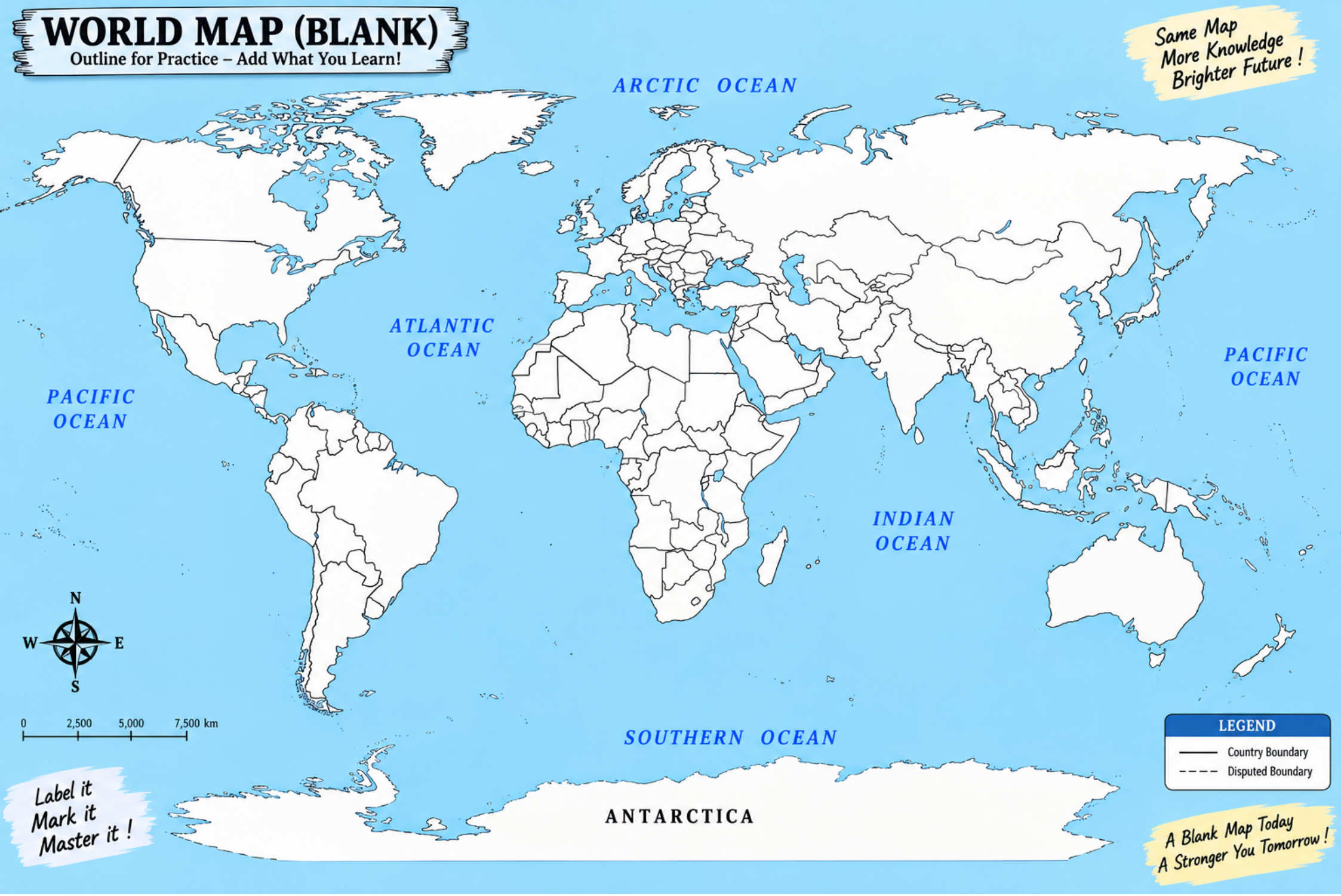
- **Gulf Stream** → North Atlantic (warm)
- **Labrador Current** → East Greenland (Cold)
- **Peru Current** → West coast of South America (cold)
- **Kuroshio Current** → Japan (warm)
- **Benguela Current** → SW Africa (cold)
- **West Australian Current** → SW Australia (cold)
- **East Australian Current** → E Australia (warm)
- **Monsoon Current** → Indian Ocean (seasonal)
- **North & South Equatorial Currents** → Pacific (warm & cold)

OCEAN CURRENTS | CLIMATE | MARINE LIFE | TRADE | SSC CGL

WORLD MAP (BLANK)

Outline for Practice – Add What You Learn!

Same Map
More Knowledge
Brighter Future !



Label it
Mark it
Master it !

A Blank Map Today
A Stronger You Tomorrow !

MAJOR SEAS OF THE WORLD

Location | Surrounded/Connected Countries | Connected Oceans | Key Facts (SSC CGL)

Seas are Smaller than Oceans, but bigger in importance for Trade, Climate and World Affairs !

Different Seas Different Lands One Connected World !

Legend (Map)

- Arctic Region
- Atlantic Region
- Indian Region
- Pacific Region
- Mediterranean Region
- Other / Inland Sea

"Different Seas Different Shores But One World We Explore !"

ARCTIC REGION

No.	Sea	Surrounded by / Connects (Countries/Lands)	Connected to (Ocean / Adjacent Water)	Key Fact (SSC CGL)
1	Barents Sea	Norway, Russia	Arctic Ocean	Major fishing area
2	Kara Sea	Russia	Arctic Ocean	Rich in natural gas
3	Laptev Sea	Russia	Arctic Ocean	One of the coldest seas
4	East Siberian Sea	Russia	Arctic Ocean	Lies between Laptev & Chukchi
5	Chukchi Sea	USA (Alaska), Russia	Arctic Ocean / Pacific	Between Bering Sea & Arctic
6	Beaufort Sea	USA (Alaska), Canada	Arctic Ocean	Important for oil & gas
7	Greenland Sea	Greenland, Norway	Arctic Ocean	Between Greenland & Svalbard
8	Norwegian Sea	Norway, Iceland	Arctic Ocean / Atlantic	Important for fisheries

ATLANTIC REGION

No.	Sea	Surrounded by / Connects (Countries/Lands)	Connected to (Ocean / Adjacent Water)	Key Fact (SSC CGL)
9	North Sea	UK, Norway, Denmark, Germany, Netherlands, Belgium.	Atlantic Ocean	Major oil & gas reserves
10	Baltic Sea	Sweden, Finland, Russia, Estonia, Latvia, Lithuania, Poland, Germany, Denmark	Atlantic (via Danish Straits)	Almost enclosed by land
11	Celtic Sea	UK, Ireland	Atlantic Ocean	Western Europe
12	English Channel	UK, France	Atlantic Ocean	Important for trade (connects to North Sea)
13	Mediterranean Sea	Europe, Africa, Asia	Atlantic (via Gibraltar)	Highly important for trade & history
14	Sargasso Sea	North Atlantic (no land boundary)	Atlantic Ocean	Surrounded by ocean currents
15	Caribbean Sea	Central America, N. South America (e.g., Mexico, Cuba, Jamaica, Hispaniola, Colombia, Venezuela etc.)	Atlantic Ocean	Important for trade & tourism
16	Gulf of Mexico	USA, Mexico, Cuba	Atlantic Ocean	Rich in oil reserves

INDIAN OCEAN REGION

No.	Sea	Surrounded by / Connects (Countries/Lands)	Connected to (Ocean / Adjacent Water)	Key Fact (SSC CGL)
17	Red Sea	Egypt, Sudan, Saudi Arabia, Yemen, Eritrea, Djibouti	Indian Ocean (via Bab-el-Mandeb)	Connects to Suez Canal
18	Arabian Sea	India, Pakistan, Iran, Oman, Yemen	Indian Ocean	Major trade route
19	Bay of Bengal	India, Bangladesh, Myanmar, Sri Lanka, Thailand	Indian Ocean	Cyclone prone region
20	Andaman Sea	India, Myanmar, Thailand, Malaysia, Indonesia	Indian Ocean	Connected to South China Sea (via Malacca Strait)
21	Laccadive Sea	India, Maldives, Sri Lanka	Indian Ocean	Known for coral reefs

PACIFIC REGION

No.	Sea	Surrounded by / Connects (Countries/Lands)	Connected to (Ocean / Adjacent Water)	Key Fact (SSC CGL)
22	Bering Sea	USA (Alaska), Russia	Pacific Ocean	Connects Asia & N. America
23	Sea of Okhotsk	Russia, Japan	Pacific Ocean	Rich in marine resources
24	Sea of Japan (East Sea)	Japan, Russia, N. & S. Korea	Pacific Ocean	Important for fisheries
25	Yellow Sea	China, N. & S. Korea	Pacific Ocean	Shallow sea
26	East China Sea	China, Japan, Taiwan	Pacific Ocean	Strategic importance
27	South China Sea	China, Vietnam, Philippines, Malaysia, Brunei, Indonesia	Pacific Ocean	Major trade route, territorial disputes
28	Philippine Sea	East of Philippines (no land boundary)	Pacific Ocean	Deepest parts of Pacific
29	Sulu Sea	Philippines, Malaysia, Indonesia	Pacific Ocean	Connects to Celebes Sea
30	Celebes Sea	Philippines, Indonesia	Pacific Ocean	Rich in biodiversity

PACIFIC REGION (contd.)

No.	Sea	Surrounded by / Connects (Countries/Lands)	Connected to (Ocean / Adjacent Water)	Key Fact (SSC CGL)
31	Java Sea	Indonesia (Java, Bornoo, Sumatra)	Pacific Ocean (via Makassar Strait)	Important for trade
32	Banda Sea	Indonesia (Banda Islands, New Guinea)	Pacific Ocean	Deep sea, rich in marine life
33	Arafura Sea	Australia, Indonesia, Papua New Guinea	Pacific Ocean	Shallow sea
34	Timor Sea	Australia, Indonesia, Timor-Leste	Pacific Ocean	Contains oil & gas reserves
35	Coral Sea	Australia, Solomon Islands, Vanuatu, New Caledonia	Pacific Ocean	World's largest coral reef (Great Barrier Reef)
36	Tasman Sea	Australia, New Zealand	Pacific Ocean	Important for shipping routes

SOUTHERN OCEAN REGION

No.	Sea	Surrounded by / Connects (Countries/Lands)	Connected to (Ocean / Adjacent Water)	Key Fact (SSC CGL)
37	Weddell Sea	Antarctica	Southern Ocean	Covered by sea ice
38	Ross Sea	Antarctica	Southern Ocean	Rich in marine life

INLAND & OTHER IMPORTANT SEAS

No.	Sea	Surrounded by / Connects (Countries/Lands)	Connected to (Ocean / Adjacent Water)	Key Fact (SSC CGL)
39	Black Sea	Turkey, Russia, Ukraine, Romania, Bulgaria, Georgia	Mediterranean (via Bosporus)	Important for trade & energy
40	Sea of Azov	Russia, Ukraine	Black Sea (via Kerch Strait)	Shallow sea
41	Caspian Sea	Russia, Karakhstan, Turkmenistan, Iran, Azerbaijan	Inland sea	World's largest inland water body

IMPORTANT STRAITS OF THE WORLD

Location | Connects | Nearby Countries | Importance (SSC CGL)

Narrow Passages Big Connections Small on Map Huge in World Affairs !

Straits Connect Continents Control Trade Shape the World !

KEY FACTS

- Over 80% of world trade by volume moves through straits and choke points.
- Straits are vital for trade, energy supplies and geopolitical control.
- Many straits are natural boundaries between continents and countries.
- They are narrow, so strategic and sometimes disputed.
- Important for SSC CGL, UPSC, State PCS and other exams.

No.	Strait	Connects	Between (Countries/Regions)	Importance / Key Fact
1	**Bering Strait**	Arctic Ocean – Pacific Ocean	Russia – USA	Separates Asia & North America
2	**Strait of Belle Isle**	Atlantic Ocean – Gulf of St. Lawrence	Canada (Newfoundland) – Labrador	Separates Canada & Newfoundland
3	**Strait of Florida**	Atlantic Ocean – Gulf of Mexico	USA – Cuba	Connects Atlantic with Gulf of Mexico
4	**Yucatán Channel**	Gulf of Mexico – Caribbean Sea	Mexico – Cuba	Important for oil & trade
5	**Mona Passage**	Atlantic Ocean – Caribbean Sea	Puerto Rico – Dominican Republic	Connects Atlantic with Caribbean
6	**Strait of Gibraltar**	Atlantic Ocean – Mediterranean Sea	Spain – Morocco	Most important European strait
7	**English Channel**	Atlantic Ocean – North Sea	UK – France	Busiest shipping route in the world
8	**Danish Strait**	Arctic Ocean – Atlantic Ocean	Greenland – Iceland	Connects Arctic with Atlantic
9	**Dardanelles**	Aegean Sea – Sea of Marmara	Turkey (European – Asian)	Important for Black Sea access
10	**Bosporus**	Sea of Marmara – Black Sea	Turkey (European – Asian)	Connects Mediterranean with Black Sea
11	**Strait of Hormuz**	Persian Gulf – Gulf of Oman	Iran – Oman (UAE nearby)	Vital for world oil trade

No.	Strait	Connects	Between (Countries/Regions)	Importance / Key Fact
12	**Rab-el-Mandeb**	Red Sea – Gulf of Aden	Yemen – Djibouti (Eritrea)	Connects Red Sea with Indian Ocean
13	**Malacca Strait**	Indian Ocean – South China Sea	Malaysia – Indonesia (Singapore)	Busiest strait in the world
14	**Sunda Strait**	Indian Ocean – Java Sea	Indonesia (Sumatra – Java)	Alternative to Malacca
15	**Lombok Strait**	Indian Ocean – Bali Sea	Indonesia (Lombok – Bali)	Used by large ships (deeper)
16	**Makassar Strait**	Java Sea – Celebes Sea	Indonesia (Borneo – Sulawesi)	Important for trade & energy
17	**Taiwan Strait**	East China Sea – South China Sea	China – Taiwan	Strategically most sensitive
18	**Korea Strait**	Sea of Japan – East China Sea	Japan – South Korea	Important for trade & naval movement
19	**Tsugaru Strait**	Sea of Japan – Pacific Ocean	Japan (Honshu – Hokkaido)	Connects Sea of Japan with Pacific
20	**La Pérouse Strait**	Sea of Japan – Sea of Okhotsk	Russia (Sakhalin) – Japan (Hokkaido)	Important for regional navigation
21	**Strait of Magellan**	Atlantic Ocean – Pacific Ocean	Chile (Mainland) – Tierra del Fuego	Alternative to Panama Canal
22	**Drake Passage**	Atlantic Ocean – Pacific Ocean	South America – Antarctica	Roughest waters in the world

QUICK REVISION TRICKS

- **Gibraltar** → Europe entry to Mediterranean.
- **Hormuz** → World oil lifeline.
- **Malacca** → Busiest strait.
- **Bering** → Separates Asia & North America.
- **Dardanelles & Bosporus** → Turkey (Black Sea).
- **Bab-el-Mandeb** → Red Sea to Indian Ocean.

"Small Straits, Global Routes."

EXAM FOCUS POINTS (SSC CGL)

- **Location** (which two water bodies does it connect)
- Countries between
- Why important (trade / oil / strategic)
- Map-based questions are commonly asked

Know the Straits Know the World !

MAJOR MOUNTAIN RANGES OF THE WORLD

Location | Highest Peak | Key Facts (SSC CGL)

Mountains Shape the World !

Different Peaks
Different People
One Earth
Many Mountains !

NORTH AMERICA

Mountain Range	Location	Highest Peak
Rocky Mountains	USA & Canada	Mount Elbert (4,401 m)
Alaska Range	Alaska (USA)	Denali (6,190 m)
Sierra Nevada	USA	Mount Whitney (4,421 m)
Cascade Range	USA	Mount Rainier (4,392 m)
Appalachian Mts.	Eastern USA & Canada	Mount Mitchell (2,037 m)

SOUTH AMERICA

Mountain Range	Location	Highest Peak
Andes	Western South America (Venezuela to Chile)	Aconcagua (6,961 m)

EUROPE

Mountain Range	Location	Highest Peak
Alps	Central Europe	Mont Blanc (4,809 m)
Scandinavian Mts.	Norway & Sweden	Galdhepiggen (2,469 m)
Ural Mountains	Europe–Asia (Russia)	Mount Narodnaya (1,895 m)

ASIA

Mountain Range	Location	Highest Peak
Himalayas	South Asia	Mount Everest (8,848 m)
Karakoram	Pakistan, China, India	K2 (8,611 m)
Hindu Kush	Afghanistan, Pakistan	(7,708 m)
Pamir	Tajikistan, China	Ismoil Somoni Peak (7,495 m)
Tien Shan	China, Kyrgyzstan, Kazakhstan	Jengish Chokusu (7,439 m)
Altai	Russia, Mongolia, China	Belukha (4,506 m)
Sayan	Russia, Mongolia	Munku-Sardyk (3,491 m)
Verkhoyansk	Russian Far East	Mount Pobeda (3,003 m)
Stanovoy	Russia	Klyuchevskaya (4,750 m)
Zagros	Iran	Zard Kuh (4,548 m)
Elburz	Iran	Mount Damavand (5,610 m)
Taurus	Turkey	Aladağlar (3,756 m)
Caucasus	Between Black & Caspian Sea	Mount Elbrus (5,642 m)

AFRICA

Mountain Range	Location	Highest Peak
Atlas Mountains	North Africa (Morocco, Algeria, Tunisia)	Jebel Toubkal (4,167 m)
Ethiopian Highlands	Ethiopia	Ras Dashen (4,550 m)
Drakensberg	South Africa, Lesotho	Thabana Ntlenyana (3,482 m)

AUSTRALIA & OCEANIA

Mountain Range	Location	Highest Peak
Great Dividing Range	Eastern Australia	Mount Kosciuszko (2,228 m)
Southern Alps	New Zealand	Aoraki / Mount Cook (3,724 m)
Maoke Range	New Zealand	(Part of Southern Alps)

IMPORTANT FACTS (SSC CGL)

- **Himalayas** → Highest, Youngest fold mountains
- **Andes** → Longest continental mountain range
- **Rockies** → Major range in North America
- **Alps** → Highest in Europe
- **Atlas** → Major range in Africa
- **Great Dividing Range** → Eastern Australia
- **Ural** → Divides Europe and Asia
- **K2** → Second highest peak in the world
- **Mt. Elbrus (Caucasus)** → Highest in Europe
- **Mount Kosciuszko** → Highest in Australia

TRICKS TO REMEMBER

- **Himalaya** → Highest
- **Andes** → America (South)
- **Rockies** → North America
- **Alps** → Europe
- **Atlas** → Africa
- **Great Dividing** → Australia
- **Ural** → Europe–Asia
- **Caucasus** → Between Black & Caspian Sea
- **Pamir** → "Roof of the World" region
- **Zagros / Elburz** → Iran

Different Mountains
Same Sky
A Higher Tomorrow !

GEOGRAPHY | WORLD | SSC CGL | LEARN SMART | SCORE HIGH

The Mountains are calling ...

SSC CGL Geography + Universe — Complete Notes

Bilkul bhai. **Map nahi — ek clean SSC CGL revision sheet** banate hain: **Dam** → **River** → **State** → **Lake/Reservoir** → **Important Fact**, flow-chart style.

■■ IMPORTANT DAMS • RIVERS • LAKES OF INDIA

SSC CGL / CHSL / MTS / Railway — Flow-Chart Revision

1■■ IMPORTANT DAMS OF INDIA

■■ DAM■ RIVER■ STATE■ RESERVOIR / LAKE■ EXAM FACT				
Tehri Dam	Bhagirathi	Uttarakhand	Tehri Reservoir	One of India's highest major dams
Bhakra Nangal	Sutlej	Himachal Pradesh/Punjab	Gobind Sagar	Major multipurpose project
Hirakud Dam	Mahanadi	Odisha	Hirakud Reservoir	Famous for very long earthen/combined dam system
Sardar Sarovar	Narmada	Gujarat	Sardar Sarovar Reservoir	Major Narmada Valley Project
Indira Sagar	Narmada	Madhya Pradesh	Indira Sagar Reservoir	Large reservoir
Omkareshwar	Narmada	Madhya Pradesh	Omkareshwar Reservoir	Narmada Valley Project
Nagarjuna Sagar	Krishna	Telangana/Andhra Pradesh	Nagarjuna Sagar Reservoir	Major Krishna basin project
Srisailam	Krishna	Andhra Pradesh/Telangana	Srisailam Reservoir	Major hydroelectric project
Tungabhadra Dam	Tungabhadra	Karnataka	Tungabhadra Reservoir	Krishna basin
Almatti Dam	Krishna	Karnataka	Almatti Reservoir	Upper Krishna Project
Koyna Dam	Koyna	Maharashtra	Shivsagar Lake	Major hydroelectric project
Jayakwadi Dam	Godavari	Maharashtra	Nath Sagar	Important irrigation project
Mettur Dam	Kaveri	Tamil Nadu	Stanley Reservoir	Major Kaveri project
Krishnarajasagara (KRS)	Kaveri	Karnataka	Krishna Raja Sagara	Near Mysuru
Idukki Dam	Periyar	Kerala	Idukki Reservoir	Famous arch dam
Mullaperiyar	Periyar	Kerala	Mullaperiyar Reservoir	Important inter-state water issue
Rihand Dam	Rihand	Uttar Pradesh	Govind Ballabh Pant Sagar	Major reservoir of UP
Gandhi Sagar	Chambal	Madhya Pradesh	Gandhi Sagar Reservoir	Chambal Valley Project
Rana Pratap Sagar	Chambal	Rajasthan	Rana Pratap Sagar	Chambal Project
Kota Barrage	Chambal	Rajasthan	—	Irrigation-focused structure
Ukai Dam	Tapi	Gujarat	Ukai Reservoir	Major Tapi basin project
Maithon Dam	Barakar	Jharkhand	Maithon Reservoir	Damodar Valley Corporation
Panchet Dam	Damodar	Jharkhand	Panchet Reservoir	Damodar Valley Project
Tilaiya Dam	Barakar	Jharkhand	Tilaiya Reservoir	First DVC dam
Konar Dam	Konar	Jharkhand	Konar Reservoir	DVC project

■ DAM → RIVER FLOW CHART

```
SUTLEJ
   ↓
Bhakra Nangal
   ↓
Gobind Sagar


BHAGIRATHI
   ↓
Tehri Dam
   ↓
Tehri Reservoir


MAHANADI
   ↓
Hirakud Dam
   ↓
Hirakud Reservoir


NARMADA
   ↓
Indira Sagar
   ↓
```

```
Omkareshwar
   ↓
Sardar Sarovar


KRISHNA
   ↓
Almatti
   ↓
Srisailam
   ↓
Nagarjuna Sagar


KAVERI
   ↓
Krishnarajasagara (KRS)
   ↓
Mettur
   ↓
Stanley Reservoir


CHAMBAL
   ↓
Gandhi Sagar
   ↓
Rana Pratap Sagar
   ↓
Kota Barrage


DAMODAR–BARAKAR SYSTEM
        ↓
Tilaiya → Konar → Maithon → Panchet
        ↓
Damodar Valley Corporation
```

2■■ IMPORTANT RIVERS OF INDIA

■■ Source■ Major States/Region■ Falls/Flows Into■ Remember				
a**	Gangotri Glacier	Uttarakhand, UP, Bihar, WB	Bay of Bengal	National river
na**	Yamunotri Glacier	Uttarakhand, Haryana, Delhi, UP	Ganga	Major tributary of Ganga
aputra**	Tibet	Arunachal Pradesh, Assam	Ganga system → Bay of Bengal	Called **Tsangpo** in Tibet
*	Tibet	Ladakh → Pakistan	Arabian Sea	Major Himalayan river
*	Tibet	HP, Punjab	Indus system	Bhakra Nangal
*	Himachal Pradesh	HP, Punjab	Sutlej	Only entirely Indian tributary of Indus
	Himachal Pradesh	HP, Punjab	Chenab	Indus system
ab**	Himachal Pradesh	HP → Pakistan	Indus system	Formed by Chandra + Bhaga
n**	Kashmir	J&K → Pakistan	Chenab	Flows through Wular Lake
ada**	Amarkantak	MP, Maharashtra, Gujarat	Arabian Sea	West-flowing
apti**	Multai, MP	MP, Maharashtra, Gujarat	Arabian Sea	West-flowing
vari**	Trimbakeshwar	Maharashtra, Telangana, AP	Bay of Bengal	**Dakshin Ganga**
a**	Mahabaleshwar	Maharashtra, Karnataka, Telangana, AP	Bay of Bengal	Peninsular river
hadi**	Chhattisgarh	Chhattisgarh, Odisha	Bay of Bengal	Hirakud
**	Talakaveri	Karnataka, Tamil Nadu	Bay of Bengal	Important South Indian river
mati**	Aravalli	Rajasthan, Gujarat	Gulf of Khambhat	West-flowing
	Aravalli	Rajasthan	Rann of Kachchh	Ends in inland drainage
	MP	MP, Rajasthan, Gujarat	Gulf of Khambhat	Crosses Tropic of Cancer
r**	Western Ghats	Kerala	Arabian Sea	Longest river of Kerala

3■■ IMPORTANT LAKES OF INDIA

■■ LAKE■ STATE/UT■ TYPE■ IMPORTANT FACT			
Wular Lake	J&K	Freshwater	One of India's largest freshwater lakes
Dal Lake	J&K	Freshwater	Srinagar; famous for houseboats
Pangong Tso	Ladakh	Brackish/saline	Transboundary lake; India–China

■■ LAKE■ STATE/UT■ TYPE■ IMPORTANT FACT			
Tso Moriri	Ladakh	High-altitude lake	Ramsar site
Chilika Lake	Odisha	Brackish lagoon	**Largest coastal lagoon in India**
Pulicat Lake	Andhra Pradesh/Tamil Nadu	Brackish lagoon	Second-largest brackish lagoon in India
Kolleru Lake	Andhra Pradesh	Freshwater	Between Krishna & Godavari deltas
Loktak Lake	Manipur	Freshwater	Famous for **Phumdis**
Sambhar Lake	Rajasthan	Saline	**Largest inland salt lake in India**
Vembanad Lake	Kerala	Brackish	Longest lake in India
Ashtamudi Lake	Kerala	Backwater	Important Ramsar site
Sasthamkotta Lake	Kerala	Freshwater	Largest freshwater lake in Kerala
Nainital Lake	Uttarakhand	Freshwater	Kumaon Himalayas
Bhojtal / Upper Lake	Madhya Pradesh	Freshwater	Bhopal
Lonar Lake	Maharashtra	Crater lake	Formed by meteorite impact
Hussain Sagar	Telangana	Artificial	Hyderabad
Nakki Lake	Rajasthan	Freshwater	Mount Abu
Umiam Lake	Meghalaya	Artificial reservoir	Near Shillong

■ LAKE → SPECIALITY FLOW CHART

```
LARGEST INLAND SALT LAKE
        ↓
     SAMBHAR
        ↓
    RAJASTHAN


LARGEST COASTAL LAGOON
        ↓
     CHILIKA
        ↓
      ODISHA


LARGEST FRESHWATER LAKE
        ↓
      WULAR
        ↓
       J&K


CRATER LAKE
     ↓
   LONAR
     ↓
 MAHARASHTRA


PHUMDIS
   ↓
LOKTAK LAKE
   ↓
MANIPUR


BETWEEN KRISHNA & GODAVARI DELTAS
              ↓
        KOLLERU LAKE
              ↓
       ANDHRA PRADESH


HOUSEBOATS
    ↓
DAL LAKE
    ↓
SRINAGAR


BRACKISH LAGOON
      ↓
```

```
CHILIKA
      ↓
ODISHA
```

■ SSC CGL MUST-REMEMBER PAIRS

DAM → RIVER

Tehri → Bhagirathi

Bhakra → Sutlej

Hirakud → Mahanadi

Sardar Sarovar → Narmada

Nagarjuna Sagar → Krishna

Mettur → Kaveri

Idukki → Periyar

Ukai → Tapi

Koyna → Koyna

Rihand → Rihand

Gandhi Sagar → Chambal

LAKE → STATE

Wular → J&K

Dal → J&K

Chilika → Odisha

Sambhar → Rajasthan

Loktak → Manipur

Kolleru → Andhra Pradesh

Pulicat → AP + Tamil Nadu

Lonar → Maharashtra

Vembanad → Kerala

Pangong Tso → Ladakh

■ SUPER TRAPS

Note: **Tehri = Bhagirathi** ■ not Alaknanda

>

Note: **Hirakud = Mahanadi**

>

Note: **Sardar Sarovar = Narmada**

>

Note: **Mettur = Kaveri**

>

Note: **Bhakra = Sutlej**

>

Note: **Idukki = Periyar**

>

Note: **Chilika = Lagoon**

>

Note: **Sambhar = Salt lake**

>

Note: **Lonar = Crater lake**

>

Note: **Loktak = Phumdis**

>

Note: **Kolleru = Krishna–Godavari region**

This is the format I'd use for your **India Geography master notes**: **Map topics separately, but Rivers–Dams–Lakes as interconnected flow charts**, because SSC frequently tests the *pairing* rather than just the individual fact.

solar sysytem adn surronding ka bano in which neigbar bhi like nebula biggest str give me notes on this

Bilkul bhai. **Solar System + uske cosmic surroundings** ko SSC CGL ke point of view se ek **complete but exam-focused master note** bana dete hain. Isme Solar System ke saath **Nebula, Galaxy, Stars, nearest neighbours, Milky Way, Local Group, Andromeda, Big Bang etc.** bhi cover honge.

■ SOLAR SYSTEM & ITS COSMIC NEIGHBOURHOOD

■ SSC CGL / CHSL / Railway — Important Notes

1■■ OUR COSMIC ADDRESS ■

```
EARTH
  ↓
SOLAR SYSTEM
  ↓
MILKY WAY GALAXY
  ↓
LOCAL GROUP
  ↓
VIRGO SUPERCLUSTER / LOCAL SUPERCLUSTER
  ↓
LARGER COSMIC STRUCTURES
  ↓
UNIVERSE
```

■ Remember

Earth → Solar System → Milky Way → Local Group → Universe

■■ 2■■ SOLAR SYSTEM

Solar System contains:

■■ 1 Sun + 8 Planets + Dwarf Planets + Moons + Asteroids + Comets + Meteoroids + Dust/Gas

Order from Sun

```
■■ SUN
 ↓
Mercury
 ↓
Venus
 ↓
Earth ■
 ↓
Mars
 ↓
Jupiter
 ↓
Saturn
 ↓
Uranus
 ↓
Neptune
```

■ Mnemonic

My Very Educated Mother Just Served Us Noodles

Mercury

Venus

Earth

Mars

Jupiter

Saturn

Uranus

Neptune

■ 3■■ PLANETS — MUST KNOW

PlanetSSC Important Fact	
Mercury	■■ Closest to Sun + smallest planet
Venus	■ Hottest planet
Earth	■ Only known planet with life
Mars	■ Red Planet
Jupiter	■ Largest planet
Saturn	■ Famous ring system
Uranus	■ Rotates almost on its side
Neptune	■ Farthest planet from Sun

■ Superlatives

```
LARGEST PLANET       → JUPITER
SMALLEST PLANET      → MERCURY
HOTTEST PLANET       → VENUS
COLDEST PLANET*      → URANUS
CLOSEST TO SUN       → MERCURY
FARTHEST FROM SUN    → NEPTUNE
RED PLANET           → MARS
RINGED PLANET        → SATURN
```

*For SSC, **Uranus** is commonly given as the coldest planet based on minimum atmospheric temperature.

■ 4■■ EARTH — IMPORTANT NUMBERS

FactValue	
Mean distance from Sun	**149.6 million km**
Rotation	**\~24 hours**
Revolution	**\~365¼ days**
Earth's natural satellite	**Moon**
Escape velocity	**11.2 km/s**
Age of Earth	**\~4.54 billion years**

Rotation → Day & Night

```
ROTATION
Earth spins on its axis
       ↓
DAY + NIGHT
```

Revolution + tilted axis → Seasons

```
REVOLUTION
     +
23.5° AXIAL TILT
     ↓
SEASONS
```

■ 5■■ MOON

Important facts

• Earth's **only natural satellite**

• No substantial atmosphere like Earth's

• No light of its own → **reflects sunlight**

• Same face generally points toward Earth due to **synchronous rotation**

• Causes **tides** mainly through gravitational attraction

• Lunar surface has **craters, maria and highlands**

Moon phases

```
New Moon
   ↓
Waxing Crescent
   ↓
First Quarter
   ↓
Waxing Gibbous
   ↓
Full Moon
   ↓
Waning Gibbous
   ↓
Last/Third Quarter
   ↓
Waning Crescent
   ↓
New Moon
```

■ 6■■ ASTEROIDS

Mostly rocky/metallic bodies orbiting the Sun.

■ Main Asteroid Belt

```
MARS
  ↓
ASTEROID BELT
  ↓
JUPITER
```

Main belt = between Mars and Jupiter

Ceres

■ **Largest object in the asteroid belt**

■ Classified as a **dwarf planet**

■■ 7■■ COMETS

Comets are mainly composed of **ice, dust and rocky material**.

When approaching the Sun:

```
ICE + DUST
    ↓
Heats up
    ↓
COMA
    ↓
TAIL
```

■ Famous Comet

Halley's Comet → approximately every 76 years

Last perihelion: **1986**

Next expected: **2061**

■■ 8■■ METEOROID → METEOR → METEORITE

Very important SSC trap:

```
METEOROID
(space)
    ↓ enters atmosphere
METEOR
("shooting star")
    ↓ reaches Earth's surface
METEORITE
```

Remember:

Space → Meteoroid

Atmosphere → Meteor

Ground → Meteorite

■ 9■■ MILKY WAY GALAXY

Our Solar System is inside the:

■ MILKY WAY

It is a **barred spiral galaxy**.

Our location

```
MILKY WAY
    ↓
ORION ARM / ORION SPUR
    ↓
SOLAR SYSTEM
```

Our Solar System lies roughly **26,000–27,000 light-years from the Galactic Centre**.

Galactic Centre

Located in the direction of the constellation **Sagittarius**.

■ ■ STARS

A **star is a huge ball of hot plasma** producing its own energy, primarily through nuclear fusion.

■■ Sun

Sun = a star

It is the **nearest star to Earth**.

■ Nearest star after Sun

Proxima Centauri

Distance ≈ **4.24 light-years**

It belongs to the **Alpha Centauri star system**.

■ 1■■1■■ BIGGEST / BRIGHTEST / NEAREST STARS

■ Biggest star — SSC level

UY Scuti is commonly cited in older competitive-exam material as one of the largest known stars.

■■ However, stellar size estimates change as observations improve, so don't treat a particular "largest star" as permanent.

■ Brightest star in night sky

Sirius

■ Nearest star to Earth

Sun

■ Nearest star after Sun

Proxima Centauri

■ North Star / Pole Star

Polaris

Located in **Ursa Minor**.

■ 1■■2■■ CONSTELLATIONS — MUST KNOW

ConstellationImportant Star / Fact	
Ursa Major	Great Bear / Saptarishi
Ursa Minor	Contains Polaris
Orion	Very prominent constellation
Cassiopeia	W-shaped constellation
Scorpius	Contains Antares
Leo	Zodiac constellation
Taurus	Zodiac constellation

■■ Indian exam connection

Saptarishi → **Ursa Major**

Dhruva Tara → **Polaris**

■■ 1■■3■■ NEBULA

This is important because you specifically asked about it.

What is a Nebula?

A **nebula** is a large cloud of **gas and dust in space**.

Nebulae can be associated with:

• ■ Star formation

• ■ Remnants of dead stars

• ■ Interstellar material

Famous Nebulae

NebulaImportant Fact	
Orion Nebula	Famous star-forming region
Crab Nebula	Supernova remnant
Eagle Nebula	Famous star-forming region
Helix Nebula	Planetary nebula
Horsehead Nebula	Dark nebula structure

■ SSC favourite

Orion Nebula → **Star formation**

■ 1■■4■■ GALAXY

A galaxy is a huge gravitationally bound system containing:

Stars + gas + dust + dark matter

Main types

```
GALAXIES
   ↓
Spiral
Elliptical
Irregular
```

Our Galaxy

Milky Way → Barred Spiral Galaxy

Nearest major galaxy

Andromeda Galaxy (M31)

■ 1■■5■■ ANDROMEDA — OUR BIG NEIGHBOUR

```
MILKY WAY
     ↕
LOCAL GROUP
     ↕
ANDROMEDA
```

Andromeda is the **nearest major galaxy to the Milky Way**.

It is approximately **2.5 million light-years away**.

■ Future event

Milky Way and Andromeda are expected to **interact and eventually merge**, on a timescale of billions of years.

■ 1■■6■■ LOCAL GROUP

The Milky Way is not alone.

It belongs to the:

LOCAL GROUP

It contains **dozens of galaxies**.

Important members

```
LOCAL GROUP
■
■■■ Milky Way
■■■ Andromeda (M31)
■■■ Triangulum (M33)
■■■ Large Magellanic Cloud
■■■ Small Magellanic Cloud
```

■ Important

Milky Way + Andromeda + Triangulum → major galaxies of Local Group

---

■■ 1■■7■■ MAGELLANIC CLOUDS

Large Magellanic Cloud

Small Magellanic Cloud

These are **dwarf satellite galaxies** associated with the Milky Way.

They are especially visible from the **Southern Hemisphere**.

■ 1■■8■■ GALAXY vs NEBULA vs STAR

This is an easy way to avoid confusion:

```
UNIVERSE
   ↓
GALAXIES
   ↓
STARS
   ↓
PLANETS
```

But **nebulae are clouds of gas/dust within galaxies**, and some are places where stars form.

Example

```
Milky Way
   ↓
Orion Nebula
   ↓
New stars forming
```

■ 1■■9■■ LIGHT-YEAR

VERY IMPORTANT

A **light-year is a unit of distance**, NOT time.

It is the distance light travels in one year.

≈ **9.46 trillion km**

Example

Proxima Centauri ≈ **4.24 light-years** away.

Meaning its light takes about **4.24 years** to reach us.

■ 2■■0■■ UNIVERSE

The universe includes:

Galaxies + stars + planets + nebulae + matter + energy + space-time

■ Big Bang Theory

The modern standard cosmological model says the observable universe evolved from an extremely hot, dense early state and has been expanding.

Important evidence

Redshift of distant galaxies → evidence for cosmic expansion

■■ 2■■1■■ BLACK HOLE

A region of spacetime with gravity so strong that **light cannot escape from within its event horizon**.

Formation

Massive stars can end their lives in supernova explosions and leave behind compact remnants; sufficiently massive remnants can form black holes.

Milky Way's centre

Sagittarius A* → supermassive black hole at the centre of the Milky Way.

■■ 2■■2■■ SUN — SSC FACTS

FactAnswer	
Sun is a	**Star**
Main composition	**Hydrogen + Helium**
Energy source	**Nuclear fusion**
Approx. age	**4.6 billion years**
Nearest star to Earth	**Sun**
Centre of	**Solar System**
Sunlight to Earth	**\~8 min 20 sec**

Fusion

```
Hydrogen
   ↓
Nuclear Fusion
   ↓
Helium + Energy
```

■ 2■■3■■ DWARF PLANETS

Important recognized dwarf planets include:

Ceres

Pluto

Haumea

Makemake

Eris

■ Trap

Pluto is NOT one of the 8 planets.

It was reclassified as a **dwarf planet in 2006**.

■■ 2■■4■■ KUIPER BELT

Located beyond Neptune.

```
NEPTUNE
   ↓
KUIPER BELT
   ↓
Trans-Neptunian Objects
```

Pluto is part of the **Kuiper Belt population**.

■ 2■■5■■ OORT CLOUD

A hypothetical/observationally inferred distant reservoir of icy bodies surrounding the Solar System.

It is thought to be the source of many **long-period comets**.

```
SOLAR SYSTEM
      ↓
KUIPER BELT
      ↓
```

```
VERY DISTANT OORT CLOUD
```

■ ULTIMATE SSC ONE-LINERS

■ Solar System

8 planets → Mercury to Neptune

Largest planet → Jupiter

Smallest planet → Mercury

Hottest → Venus

Closest to Sun → Mercury

Farthest planet → Neptune

Red Planet → Mars

Planet famous for rings → Saturn

Earth's satellite → Moon

Largest dwarf planet → Pluto *(by commonly used exam convention; Eris is more massive)*

■ Stars

Nearest star → Sun

Nearest star after Sun → Proxima Centauri

Brightest night-sky star → Sirius

Pole Star → Polaris

Sun → Star

Largest star often quoted in exams → UY Scuti

■ Galaxies

Our galaxy → Milky Way

Milky Way type → Barred Spiral

Nearest major galaxy → Andromeda

Andromeda → M31

Milky Way + Andromeda → Local Group

■ Nebula

Cloud of gas and dust → Nebula

Star formation → Orion Nebula

Supernova remnant → Crab Nebula

■ Space Objects

```
ASTEROID
→ Rocky/metallic body

COMET
→ Ice + dust + rock

METEOROID
→ Small body in space

METEOR
→ Burns/glows in atmosphere

METEORITE
```

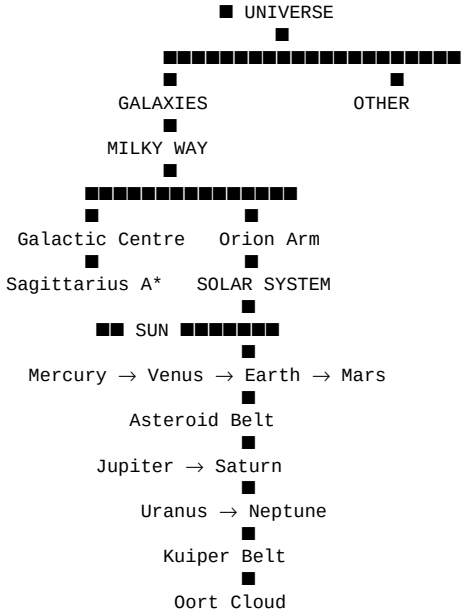
```
→ Reaches Earth's surface

NEBULA
→ Gas + dust cloud

GALAXY
→ Huge gravitationally bound system

BLACK HOLE
→ Extremely strong gravitational region
```

■■ YOUR COSMIC MAP — REMEMBER THIS



■ The 15 facts I would memorise first for SSC

1. Solar System → **8 planets**

2. Largest planet → **Jupiter**

3. Smallest → **Mercury**

4. Hottest → **Venus**

5. Farthest planet → **Neptune**

6. Red Planet → **Mars**

7. Our galaxy → **Milky Way**

8. Milky Way → **Barred spiral**

9. Nearest major galaxy → **Andromeda (M31)**

10. Our galaxy's group → **Local Group**

11. Nearest star → **Sun**

12. Nearest star after Sun → **Proxima Centauri**

13. Brightest night-sky star → **Sirius**

14. Gas + dust cloud → **Nebula**

15. Light-year → **Unit of distance**

Ye topic tumhare previous Geography posters ke saath Astronomy/Universe section ka strong base ban jayega.

unvirde

make a html i everthing in this

chat posterds notes

not a single should be missing